

# INVARIANT ARITHMETIC OF CLOSED GEODESICS

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**ABSTRACT.** We study rational valued counts of geodesic strings (reparametrization equivalence classes of closed geodesics) for complete Riemann-Finsler manifolds, based on the Fuller index of the geodesic flow. The main conceptual result is a product type formula for these counts. Combined with KAM theory, it yields the following new phenomenon. Let  $g$  be a generic Finsler metric on  $T^2$ ,  $C^\aleph$ - $\hbar$ -close to a flat metric, with  $\hbar > 0$  and  $\aleph < \infty$  some effective constants. Let  $\beta_0$  be an indivisible free homotopy class, and let  $e$  be odd. If  $g$  has a prime geodesic string in the class  $e\beta_0$ , then it has another one. This is one half of a complete arithmetic law satisfied by the signed counts of prime geodesic strings in the classes  $e\beta_0$ , which in the even classes takes the form of a period doubling constraint. An analogous law holds for metrics  $C^1$ -close to a negative curvature metric on a closed orientable surface, with Gromov hyperbolicity replacing KAM. We also obtain arithmetic constraints on counts of closed geodesics in mapping tori and flat bundles, and constraints on the existence of negative sectional curvature metrics. Finally, we give evidence for a topological invariance phenomenon of these counts; such invariance would give analogous arithmetic laws for *all* metrics on manifolds having the tautness property studied here.

## 1. INTRODUCTION

A *geodesic string* will be a short name for the equivalence class of a closed, constant speed  $g$ -geodesic up to reparametrization  $S^1$  action, we say it is *prime* if it does not have a multiply covered representative. More generally, we say that a smooth loop is *prime* if it is not multiply covered. Suppose for the moment  $X$  is compact. For a nontrivial free homotopy class  $\beta$  of loops in  $X$ , we say that a metric  $g$  on  $X$  is  $\beta$ -*regular* if all of its class  $\beta$  closed geodesics are nondegenerate in the usual sense, or equivalently the closed orbits of the associated geodesic flow are dynamically nondegenerate.

We will study arithmetic properties of geodesic string counts, which are outside the scope of ( $S^1$ -equivariant) Morse homology of the loop space. A principal tool is a product formula, Section 7, for this count and its generalizations based on Fuller index theory. The Fuller index is a completely general construction in dynamical systems; the product formula is a more subtle phenomenon, and needs the Riemann-Finsler geometric context. Fuller's theory and the basic definitions are outlined in the Appendix, so no prior knowledge is necessary.

The constants  $\hbar > 0$ ,  $\aleph \in \mathbb{N}$  below will be fixed throughout the paper, instantiated in Section 10, and are effective, that is, in principle computable. The product formula, combined with KAM confinement near flat metrics and Gromov hyperbolicity near negative curvature, gives:

**Theorem 1.1** (Proof in Section 11). *Let  $X$  be a closed orientable surface,  $\beta_0$  an indivisible free homotopy class, and  $g$  an  $e\beta_0$ -regular metric on  $X$ , for a given  $e \geq 1$ . Suppose one of the following holds:*

- $X = T^2$  and  $g$  is  $C^\aleph$ - $\hbar$ -close to a flat metric  $g_0$ , that is  $\|g - g_0\|_{C^\aleph} < \hbar$  (with the norm convention of Section 10);
- $g$  is  $C^1$ - $\epsilon$ -close to a metric  $g_-$  of negative curvature on  $X$ , with  $\epsilon = \epsilon(g_-)$  as in Proposition 2.7. Such a  $g$  can have pointwise arbitrarily large curvature, see Example 4.

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Then:

- (1) For  $e$  odd, if there is a prime geodesic string in class  $e\beta_0$ , there is another — except for  $e = 1$  in the negative curvature case, where the number of prime geodesic strings in class  $\beta_0$  is instead odd.
- (2) For  $e = 2n$ , the class  $2n\beta_0$  contains at least as many prime geodesic strings as the class  $n\beta_0$  contains inverse-hyperbolic ones (the linearized return map has negative eigenvalues) — a period doubling phenomenon.

Note that  $e$  odd is essential in part (1): for  $e$  even a counterexample is given in Example 2 below. The prime geodesic strings in the hypotheses are indeed realized, in both settings, by Hill type engineering as in Example 2. In the case of  $g$   $C^1$ -close to  $g_-$ , this engineering demands that  $\|g - g_-\|_{C^2}$  be large: otherwise we stay in the negative curvature regime. In the negative curvature case, part (1), including its  $e = 1$  variant, is not surface-specific: it holds on closed orientable manifolds of any dimension, by Proposition 2.7, Example 5 and Theorem 1.9. For  $e = 1$  the  $T^2$  pairing holds unconditionally. For every metric on  $T^2$ : an indivisible class contains a simple closed geodesic, and Lusternik–Schnirelmann min-max of the length over a sweepout family of the class, run through the curve shortening flow [24], produces either a second simple closed geodesic of greater length, or infinitely many of the same length. I thank Marco Mazzucchelli for telling me about this.

The above is obtained from invariance properties of the following count. For a  $\beta$ -regular  $g$  with finitely many class  $\beta$  geodesic strings, we define:

$$(1.2) \quad F(g, \beta) = \sum_o \frac{(-1)^{\text{morse}(o)}}{\text{mult}(o)} \in \mathbb{Q},$$

where:

- The sum is over class  $\beta$   $g$ -geodesic strings  $o$ .
- $\text{morse}(o)$  denotes the Morse index of  $o$  (meaning the Morse-Bott index of the associated critical submanifold (diffeomorphic to  $S^1$ ) of the loop space).
- $\text{mult}(o)$  is the geometric multiplicity, equivalently the order of the corresponding isotropy subgroup of  $S^1$ , where  $S^1$  is acting on the loop space by reparametrization.

For a free homotopy class  $\beta$  of loops in  $X$ , let  $L_\beta X$  denote the corresponding component of the free loop space. The following is a direct corollary of Theorem 2.17, and exemplifies the basic topological invariance of the  $F$ -count:

**Corollary 1.3.** *Let  $X$  be compact,  $g, \beta$  as above with finitely many class  $\beta$   $g$ -geodesic strings, and such that  $\beta$  is indivisible (cannot be represented by a multiply covered loop). Then the  $S^1$ -equivariant homology  $H_*^{S^1}(L_\beta X, \mathbb{Z})$  has finite total rank. Denote by  $\chi^{S^1}(L_\beta X)$  the Euler characteristic of this homology. Then*

$$F(g, \beta) = \sum_o (-1)^{\text{morse}(o)} = \chi^{S^1}(L_\beta X),$$

(note  $\text{mult}(o) = 1$  by the indivisible condition.)

The proof of the above is based on  $S^1$ -equivariant Morse theory. For divisible classes, the homology theoretic interpretation of the  $F$ -count would require more than ordinary  $S^1$ -equivariant Morse homology for the  $g$ -energy functional. One would want a  $g$ -invariant up to quasi-isomorphism “orbifold Morse homology chain complex” for the quotient stack  $[LX/S^1]$ , together with (assuming certain finiteness) an algebraically derived  $\mathbb{Q}$ -valued characteristic<sup>1</sup> coinciding with the  $F$ -count. This is likely related to Wall-Euler characteristic theory in group cohomology theory, which also appears in the

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<sup>1</sup>I am told that in this context a better term is von Neumann dimension.

more basic Conjecture 1. Existence of this orbifold Morse homology is at the moment unknown; for a finite dimensional candidate see [20].

**1.1. Tautness condition.** For a compact  $X$ , we say a metric  $g$  is **taut** if the space of class  $\beta$  closed  $g$ -geodesics is compact for every  $\beta$ . (With the topology induced by the  $C^1$  topology on the loop space; on the space of closed geodesics this agrees with the compact open topology of the formal Definition 2.2.) It is said to be  **$\beta$ -taut** if the same holds for just  $\beta$ -class geodesics. A homotopy  $\{g_t\}_{t \in [0,1]}$  is said to be  **$\beta$ -taut** if the space of pairs  $(o, t)$  with  $o$  class  $\beta$   $g_t$ -geodesic is compact. Examples of metrics  $\beta$ -taut homotopic to a flat metric, arbitrarily “far” from flat, are supplied by Proposition 1.4 below. If  $X$  admits a negative curvature metric, then there are  $C^1$  open sets of taut metrics on  $X$ , containing metrics of arbitrarily large curvature, all taut homotopic to a negative curvature metric. See Example 4.

**Proposition 1.4** (Clairaut metrics on  $T^n$ ; proof in Section 12). *Write  $T^n = T^{n-1} \times S^1$ , with coordinates  $(x, y)$ , and let*

$$g = dy^2 + \sum_{i,j} a_{ij}(y) dx_i dx_j,$$

*for  $a(y)$  symmetric positive definite, so  $g$  is invariant under the  $T^{n-1}$  action. Let  $\beta = (m, 0)$ ,  $m \neq 0$ , be a fiber class. Then  $g$  is  $\beta$ -taut,  $\beta$ -taut homotopic to a flat metric, and  $F(g, \beta) = 0$ , with  $F$  as in Theorem 2.8.*

For  $n \geq 3$  there is no KAM confinement (Remark 10.6), but tautness abounds: the proposition gives far from flat taut metrics, with  $F = 0$  on the fiber class rays, in every dimension. Since  $\pi_1(T^n)$  is free abelian, Theorem 1.9 then gives the odd class law  $s_e = 0$  for any  $e\beta$ -regular metric joined to a Clairaut metric by an  $e\beta$ -taut deformation (its divisor levels are taut automatically, as covers of strings are strings). The Clairaut metrics themselves are not regular for  $n \geq 3$ : their parallels come in  $T^{n-1}$ -families of geodesic strings.

In the other direction,  $\beta$ -tautness fails easily.

*Example 1.* Let  $g$  agree with a flat metric outside a small embedded disk in  $T^2$ , inside which it is rotationally symmetric with a “mushroom” profile: a bulb over a neck, the circumference having a strict local minimum at a waist circle  $\gamma_0$ , which is then a closed geodesic. Let  $S \subset T^2$  be the complement of the open bulb: a compact surface with geodesic boundary  $\gamma_0$ , with  $\pi_1(S)$  free on  $x, y$  and  $[x, y]$  the boundary class, killed by the inclusion into  $T^2$ . For  $k \in \mathbb{N}$  the elements  $x[x, y]^k$  represent pairwise distinct nontrivial conjugacy classes of  $\pi_1(S)$  (their cyclically reduced lengths differ), all mapping to the class  $\beta$  of  $x$  in  $T^2$ . As  $\partial S$  is geodesic, minimizing length in each of these classes produces closed  $g$ -geodesics  $o_k$  of class  $\beta$  in  $T^2$ ; and since for each  $L$  only finitely many free homotopy classes of the compact  $S$  contain a loop of length  $\leq L$ ,  $\ell_g(o_k) \rightarrow \infty$ . So  $g$  is not  $\beta$ -taut, and the argument applies to every nontrivial class. The disk may be taken arbitrarily small, so the failure is not curable by localizing the perturbation: the  $C^{k_0}$ -closeness in Proposition 10.1 ahead is essential.

Using geodesic confinement based on KAM theory, and the product formula of Theorem 7.1 we show in Section 10 the following:

**Theorem 1.5.** *Any Finsler metric  $g$  on  $T^2$ ,  $C^\infty$ - $\hbar$ -close to a flat metric, is taut homotopic to that flat metric.*

We will show that the count  $F(g, \beta) \in \mathbb{Q}$  can be defined for any  $\beta$ -taut  $g$  (Section 5). In fact, near the flat metrics the invariant vanishes identically, and the resulting arithmetic constraints are how one gets Theorem 1.1.

**Theorem 1.6** (Proof in Section 11). *Let  $g$  be a Riemann-Finsler metric on  $T^2$ ,  $C^\infty$ - $h$ -close to a flat metric. Then for every nontrivial class  $\beta$ ,  $g$  is  $\beta$ -taut and*

$$F(g, \beta) = 0.$$

*Question 1.* Let  $g$  be an  $e\beta_0$ -taut metric on  $T^2$ , for an indivisible class  $\beta_0$  and  $e$  odd. If the class  $e\beta_0$  contains a prime closed geodesic string, must it contain at least two?

“Yes” does not follow formally from the vanishing of  $F$ : a single degenerate prime string, of vanishing local index (Appendix A), is consistent with  $F(g, e\beta_0) = 0$ . It might follow from some min-max theory of the  $S^1$ -equivariant pair  $(L_{e\beta_0}T^2, I)$ , with  $I$  the locus of iterated strings — the locus of nontrivial isotropy — in the spirit of the Lusternik–Schnirelmann argument above.

The near flat case of Theorem 1.1 generalizes to any taut regular metric with vanishing  $F$  on the divisor classes, see Proposition 11.5. In this case the inverse-hyperbolic type is realized, by the following example, which moreover has exactly three nondegenerate class  $2\beta_0$  geodesic strings.

*Example 2.* Take  $T^2 = \mathbb{R}^2/\mathbb{Z}^2$  and let  $\beta_0$  be the indivisible class of the circle  $\{y = 0\}$ . We exhibit a metric whose class  $\beta_0$  geodesic strings are exactly two: the minimizer  $\{y = 1/2\}$ , of length  $1/2$  and positive hyperbolic type, and  $o = \{y = 0\}$ , of length 1 and inverse-hyperbolic type. The class  $2\beta_0$  geodesic strings are exactly three: the two double covers together with a single prime geodesic string  $\gamma$ , of elliptic type, born of  $o$  at a period doubling. See Figure 1. So  $s_1 = 0$ ,  $b_1 = 1$  and  $s_2 = -1 = -b_1$ : the law (11.6) is realized with a single prime string, attaining the bound of Theorem 1.1, and

$$F(g, \beta_0) = 1 - 1 = 0, \quad F(g, 2\beta_0) = \frac{1}{2} + \frac{1}{2} - 1 = 0.$$

In Fermi form  $g = dy^2 + A(x, y)^2 dx^2$ , the curve  $\{y = 0\}$  is a unit speed closed geodesic of length 1 as soon as  $A(x, 0) \equiv 1$  and  $\partial_y A(x, 0) \equiv 0$ ; the Gauss curvature of  $g$  along this curve is then the function  $K(x) = -\partial_y^2 A(x, 0)$  of the arc parameter  $x$ , and the normal variational equation is the Hill equation

$$J'' + KJ = 0.$$

We then take:

$$A(x, y) = 1 - \frac{1}{2} \left( 1 + \frac{5}{\pi^2} \cos 2\pi x \right) \sin^2 \pi y.$$

The full details of the example are in Section 12.

Theorem 1.5 settles the following question near the flat metric.

*Question 2.* Is every  $\beta$ -taut metric on  $T^2$   $\beta$ -taut homotopic to the flat metric?

$T^n$ , and  $T^2$  especially, have a concrete universe of taut metrics (near flat, Clairaut type), as does any closed orientable manifold with negative curvature ( $C^1$ -balls, Proposition 2.7). Moving beyond these universes we have the following.

**Definition 1.7.** An indivisible class  $\beta \in \pi_1^{\text{inc}}(X)$  of a manifold  $X$  satisfies the **cancellation property** if for all  $k, e \geq 1$  and every class  $\gamma$ ,  $k\gamma = e\beta$  implies  $k|e$  and  $\gamma = (e/k)\beta$ .

**Lemma 1.8** (Proof in Section 11). *Every indivisible class of  $X$  satisfies the cancellation property when  $\pi_1(X)$  is free abelian, or torsion-free hyperbolic.*

By this lemma the following theorem applies to  $T^n$  or a closed orientable manifold admitting a negative curvature metric.

$\{y = 1/2\}$ : the minimizer,  $\ell = 1/2$ , positive hyperbolic

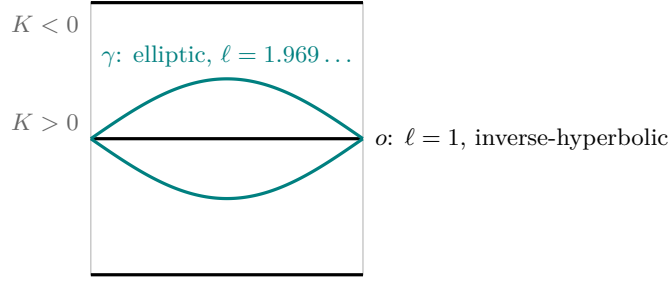


FIGURE 1. Schematic of Example 2 on the fundamental domain  $[0, 1] \times [-\frac{1}{2}, \frac{1}{2}]$ , sides identified. The two teal strands form the prime geodesic string  $\gamma$  of class  $2\beta_0$ : it winds twice in  $x$  while oscillating once about  $o$ , born of  $o$  at a period doubling. Lengths are with respect to  $g$ , the picture is topological.

**Theorem 1.9** (Proof in Section 11). *Let  $X$  be a closed orientable manifold,  $\beta$  an indivisible class, and  $e \geq 1$  odd. Let  $g$  be an  $e\beta$ -taut and  $e\beta$ -regular metric on  $X$ , and suppose that  $\beta$  satisfies the cancellation property. Let  $P_d$  be the set of prime geodesic strings of class  $d\beta$ , and  $s_d := \sum_{o \in P_d} i(o)$  its signed count, with  $i$  the local Fuller index (A.2) — defined, as regularity of the divisor classes is automatic (Section 11). Then*

$$(1.10) \quad e F(g, e\beta) = \sum_{d|e} d s_d,$$

and hence, by Möbius inversion,

$$(1.11) \quad s_e = \frac{1}{e} \sum_{d|e} \mu(e/d) d F(g, d\beta),$$

with  $\mu$  the Möbius function. In particular,

- (1) If  $F(g, d\beta) = 0$  for every  $d|e$  then  $s_e = 0$ , and if the class  $e\beta$  contains a prime geodesic string, it contains another.
- (2) If  $e > 1$  and  $F(g, d\beta) = 1/d$  for every  $d|e$  then  $s_e = 0$ , and if the class  $e\beta$  contains a prime geodesic string, it contains another. For  $e = 1$ , (1.10) reads  $s_1 = F(g, \beta) = 1$ : the class  $\beta$  contains an odd number of prime geodesic strings.

*Remark 1.12.* The second alternative is the negative curvature pattern: there  $F(g, d\beta) = 1/d$  by Example 5. Unconditionally, Proposition 2.7 provides it on the  $C^1$ -balls about negative curvature metrics. Combined with Conjecture 1 just below, it would apply to every  $e\beta$ -taut and  $e\beta$ -regular metric on a closed orientable manifold admitting a negative curvature metric — the cancellation property holds there unconditionally, as  $\pi_1(X)$  is torsion-free hyperbolic (Lemma 1.8).

**1.2. The topology connection.** What follows is a special version of Conjecture 2 (appearing further ahead) and can be understood as a divisible analogue of Corollary 1.3 in the aspherical case.

**Conjecture 1** (Aspherical topological formula for  $F$ ). *Let  $X$  be a closed aspherical manifold, and let  $\beta$  be a nontrivial free homotopy class. Suppose that  $X$  admits a  $\beta$ -taut metric. Then the Borel construction*

$$(L_\beta X)_{hS^1} := L_\beta X \times_{S^1} ES^1$$

*is Wall-Euler finite, and for every  $\beta$ -taut metric  $g$ ,*

$$F(g, \beta) = \chi_{\text{Wall}}((L_\beta X)_{hS^1}).$$

Equivalently, set  $G = \pi_1(X)$ , let  $C_G(\gamma_\beta)$  be the centralizer of  $\gamma_\beta$  in  $G$  and  $Q_\beta = C_G(\gamma_\beta)/\langle \gamma_\beta \rangle$ , where  $\beta = [\gamma_\beta]$  is the conjugacy class of  $\gamma_\beta \in G$ . Then  $Q_\beta$  is Wall-Euler finite and

$$F(g, \beta) = \chi_{\text{Wall}}(Q_\beta).$$

The conjecture implies that  $F(g, \beta)$  is independent of the  $\beta$ -taut metric  $g$ , for closed aspherical manifolds. For background on rational Euler characteristics of groups, we refer to Wall's original note [53] and to Brown's treatment of Euler characteristics of groups in [18, Chapter IX]; see also Brown [17] for Euler characteristics of discrete groups and  $G$ -spaces.

Note that if we assume the above conjecture, then  $F(g, \beta) = 0$  for every  $\beta$ -taut metric  $g$  on  $T^n$ , so that the conclusions of Proposition 11.5 hold for any suitably regular  $g$  without a near flat hypothesis.

The following is a rather important open question in our context:

*Question 3.* Suppose that  $X$  is closed, if  $X$  admits a  $\beta$ -taut metric for some nontrivial  $\beta$ , must  $X$  be aspherical?

*Remark 1.13.* A possible route to a counterexample for Question 3 is a mapping torus of a non-aspherical  $Z$  by a diffeomorphism with “hyperbolic behavior”. In the other direction, asphericity is not sufficient: by results of Mess [39] and Belegarde [9, Corollary 5.1], in every dimension  $n \geq 4$  there are closed aspherical  $n$ -manifolds whose fundamental groups contain nontrivial infinitely divisible elements, and hence, by Proposition 1.14 below, classes  $\beta$  for which no  $\beta$ -taut metric exists.

**Proposition 1.14.** *Let  $X$  be a closed manifold and let  $\beta \neq 1$  be a free homotopy class which is infinitely divisible in  $\pi_1(X)$ : for arbitrarily large  $n \in \mathbb{N}$ , a representative of  $\beta$  in  $\pi_1(X)$  is an  $n$ -th power. Then  $X$  admits no  $\beta$ -taut metric.*

*Proof.* Take any metric  $g$  and let  $l > 0$  be the systole of  $g$ , that is

$$l = \inf_{\{o \mid o \text{ is noncontractible}\}} \ell_g(o).$$

For  $n$  as in the statement, write a representative of  $\beta$  as  $h_n^n$ , with  $h_n$  necessarily noncontractible since  $\beta \neq 1$ . A length-minimizing closed geodesic in the free homotopy class of  $h_n$  exists as  $X$  is closed, and its  $n$ -fold cover is a class  $\beta$  closed  $g$ -geodesic of length at least  $n \cdot l$ . So the class  $\beta$  closed geodesics of  $g$  have unbounded length and  $g$  is not  $\beta$ -taut.  $\square$

**1.3. Geodesics in mapping tori and negative sectional curvature.** A theorem of Preissmann [47] says that there are no negative sectional curvature metrics on compact products. For fibrations, the most obvious analogue of Preissmann's theorem fails even assuming compactness. By Thurston's fibered hyperbolization theorem, a closed surface bundle over  $S^1$  with fiber of negative Euler characteristic is hyperbolic if and only if its monodromy is pseudo-Anosov; see [52, 51]. What follows is a kind of partial, “only if” Finsler analogue in higher dimensions, noting that an isometry of a manifold has “anti-Anosov behavior” (for some reasonable generalization of pseudo-Anosov to higher dimensions).

We denote by  $\pi_1^{\text{inc}}(X)$  the set of free homotopy classes of loops incompressible to the ends, in the sense of Definition 2.1, (by convention nonconstant classes when  $X$  is closed).

**Corollary 1.15** (Corollary of Corollary 2.28). *Let  $X_\phi$  be the mapping torus of an isometry  $\phi : Z \rightarrow Z$  of a complete taut Riemannian metric on  $Z$ , with  $Z$  finite type, and where  $\beta_Z \in \pi_1^{\text{inc}}(Z)$  is indivisible. Let  $\beta \in \pi_1^{\text{inc}}(X_\phi)$  denote the image of  $\beta_Z$  under inclusion. Then:*

$$(1) \quad \chi^{S^1}(L_\beta X_\phi) = 0.$$

- (2)  $X_\phi$  does not admit a forward complete,  $\beta$ -regular Riemann-Finsler metric with an odd number of class  $\beta$  geodesic strings.
- (3)  $X_\phi$  does not admit a complete negative sectional curvature Riemannian metric.

*Example 3.* The corollary applies to the mapping torus of a finite type surface of genus at least one (the genus condition allows us to find a nontrivial class in  $\pi_1^{inc}(X_\phi)$ , we also need a suitable taut metric e.g. a nonpositive curvature metric, with an isometry). The conclusion that  $X_\phi$  does not admit a complete negative sectional curvature metric follows from Thurston's classification<sup>2</sup>. However the arithmetic constraints on geodesic counts seem interesting even for a surface map.

If we further assume that  $Z$  is aspherical, the first part of Corollary 1.15 can be proved via group cohomology theory, for which we already provided some references. But the main point is that in our setup the proof is elementary, once the general product formula of Theorem 7.1 for the  $F$ -count is established. The proof of the latter does involve a novel type of perturbation, but Fuller index theory allows us to do this abstractly, perturbing a vector field rather than the metric/energy functional.

**Organization of the paper.** Section 2 sets up geodesic strings, taut metrics and taut deformations, states the supporting theorems, and contains the applications to sky catastrophes. Section 3 proves the compactness results underlying tautness, in particular Theorem 2.5. Section 4 collects Reeb-flow preliminaries, and Section 5 constructs the invariant  $F$  and proves Theorem 2.8. Section 6 proves the Euler characteristic formula for indivisible classes, and Sections 7 and 8 prove the product and fibration formulas. Section 9 proves the  $C^1$  stability Proposition 2.7 via Gromov hyperbolicity, Section 10 proves the near flat rigidity Proposition 10.1 by KAM confinement, and Section 11 proves the vanishing Theorem 1.6, Theorem 1.9 and the arithmetic law of Proposition 11.5. Section 12 proves the Clairaut Proposition 1.4. Appendix A develops the Fuller index and its continuation theory in the noncompact setting, Appendix B performs a local fibered computation of  $F$  used in the product formula, and Appendix C contains the facts about sky catastrophes used in Section 2.0.2.

## 2. SETUP AND SUPPORTING THEOREMS

Proofs of many results here are postponed till subsequent sections.

**Terminology 1.** From now on, all our metrics are Riemann-Finsler (a.k.a. Finsler) metrics unless specified to be Riemannian, and usually denoted by just  $g$ . Completeness, always means forward completeness, and it is an assumption for all our metrics. Curvature always means sectional curvature in the Riemannian case and flag curvature in the Finsler case. Thus we will usually just say complete metric  $g$ , for a forward complete Riemann-Finsler metric. A reader may certainly choose to interpret all metrics as Riemannian metrics, completeness as standard completeness, and curvature as sectional curvature.

In what follows,  $\pi_k(X)$  denotes the usual based homotopy group, with basepoints suppressed when no confusion can arise.

**Definition 2.1.** Let  $X$  be a smooth manifold. Fix an exhaustion by nested compact sets  $\bigcup_{i \in \mathbb{N}} K_i = X$ ,  $K_i \supset K_{i-1}^\circ$  for all  $i \geq 1$ . A free homotopy class  $\beta$  of loops in  $X$  is called **end compressible** if, for every  $i$ , it has a representative contained in  $X - K_i$ . We say that  $\beta$  is **end incompressible** (or incompressible to the ends) if it is not end compressible.

Let  $\pi_1^{inc}(X)$  denote the set of such end incompressible free homotopy classes. When  $X$  is compact, we set  $\pi_1^{inc}(X)$  to be the set of nonconstant free homotopy classes of loops in  $X$ .

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<sup>2</sup>Suitably adjusted, one may also need finite geometry (volume) assumptions.

It is easily seen that the above is well defined (independent of the choice of an exhaustion) and moreover any homeomorphism  $X_1 \rightarrow X_2$  of a pair of manifolds induces a set isomorphism  $\pi_1^{inc}(X_1) \rightarrow \pi_1^{inc}(X_2)$ .

Denote by  $L_\beta X$  the class  $\beta \in \pi_1^{inc}(X)$  component of the free loop space of  $X$ , with its compact open topology. Let  $g$  be a complete metric on  $X$ , and let  $S(g, \beta) \subset L_\beta X$  denote the subspace of all constant speed parametrized, smooth, closed  $g$ -geodesics in class  $\beta$ .

**Definition 2.2.** We say that a metric  $g$  on  $X$  is  $\beta$ -**taut** if it is complete and  $S(g, \beta)$  is compact. We will say that  $g$  is **taut** if it is  $\beta$ -taut for each  $\beta \in \pi_1^{inc}(X)$ .

**Lemma 2.3.** *A complete metric  $g$  with nonpositive curvature satisfies:*

- *All of its closed geodesics are minimizing in their free homotopy class.*
- *It is taut.*

*Proof.* The first part is a standard consequence of the Cartan-Hadamard theorem. The second part follows by the first part and Corollary 3.3.  $\square$

**Definition 2.4.** Let  $\beta \in \pi_1^{inc}(X)$ , and let  $g_0, g_1$  be a pair of  $\beta$ -taut metrics on  $X$ . A  $\beta$ -**taut deformation** between  $g_0, g_1$ , is a continuous (in the topology of  $C^0$  convergence on compact sets) family  $\{g_t\}$ ,  $t \in [0, 1]$  of complete metrics on  $X$ , such that

$$S(\{g_t\}, \beta) := \{(o, t) \in L_\beta X \times [0, 1] \mid o \in S(g_t, \beta)\}$$

is compact. We also say in this case that  $g_i$  are  $\beta$ -taut homotopic.<sup>3</sup> We say that  $\{g_t\}$  is a **taut deformation** if it is  $\beta$ -taut for each  $\beta \in \pi_1^{inc}(X)$ . The above definitions of tautness are extended naturally to the case of a smooth fibration  $X \hookrightarrow P \rightarrow [0, 1]$ , with a smooth fiber-wise family of metrics.

A useful criterion for  $\beta$ -tautness is the following.

**Theorem 2.5** (Proof in Section 3.1). *Let  $\beta \in \pi_1^{inc}(X)$ . Let  $\{g_t\}_{t \in [0, 1]}$  be a continuous family of complete metrics on  $X$ . Suppose that:*

$$\sup_t \left| \sup_{o \in S(g_t, \beta)} \ell_{g_t}(o) - \inf_{o \in S(g_t, \beta)} \ell_{g_t}(o) \right| < \infty,$$

*where  $\ell_{g_t}$  is the length functional with respect to  $g_t$ , then  $\{g_t\}$  is  $\beta$ -taut.*

For example, the hypothesis is trivially satisfied if  $g_t$  have the property that all their class  $\beta$  closed geodesics are minimizing in their homotopy class.

**Corollary 2.6.** *If  $g_t$ ,  $t \in [0, 1]$  have nonpositive curvature then  $\{g_t\}$  is taut.*

*Proof.* This follows by Theorem 2.5 and by Lemma 2.3.  $\square$

The following supplies more examples of tautness, and the negative curvature case of Theorem 1.1: strictly negative curvature gives tautness an open  $C^1$  neighborhood in the space of all metrics.

**Proposition 2.7** ( $C^1$  stability of tautness in negative curvature; proof in Section 9). *Let  $g_-$  be a metric of negative sectional curvature on a closed manifold  $X$ . There is  $\epsilon = \epsilon(g_-) > 0$  such that every metric  $g'$  with  $\|g' - g_-\|_{C^1} < \epsilon$  is taut. Any two metrics in this ball are joined by a taut deformation. In particular  $F(g', \beta) = F(g_-, \beta)$  for every nontrivial  $\beta$ , with  $F$  as in Theorem 2.8.*

<sup>3</sup>Let us point out that this does not necessarily hold if  $g_i$  are path connected by a path of  $\beta$ -taut metrics.

$C^1$ -smallness permits arbitrarily large curvature.

*Example 4.* On a closed hyperbolic surface  $(\Sigma, g_-)$  take  $g' = e^{2u}g_-$ , with  $u = \delta w \phi(d_{g_-}(\cdot, x_0)/w)$  for a fixed bump function  $\phi$  with a nondegenerate maximum at 0. Then  $\|u\|_{C^1} \leq C\delta$ , while  $K(g') = e^{-2u}(-1 - \Delta u)$  is of order  $\delta/w$  at  $x_0$ . For  $w$  small this produces metrics in the ball of Proposition 2.7 with curvature maxima as large as desired — so  $C^2$ -far from every metric of nonpositive curvature — and any number of such bumps may be placed. The same bumps on a negatively curved  $(X, g_-)$  of any dimension give sectional curvature of order  $\delta/w$  in every plane at  $x_0$ , by the conformal curvature formula at a critical point of  $u$ . These metrics are taut with  $F(g', \beta) = 1/n$ , for  $\beta$  represented by a multiplicity  $n$  closed  $g_-$ -geodesic, by Proposition 2.7 and Example 5. In particular the  $1/d$  ray values of Remark 1.12 hold for them unconditionally, and Theorems 1.9 and 1.1 apply to their  $e\beta$ -regular members. The mushroom of Example 1, planted on  $\Sigma$ , shows  $C^0$ -smallness does not suffice.

2.0.1. *The geodesic string counting invariant  $F$ .* Let  $\mathcal{G}(X)$  be the set of complete metrics on  $X$ . The following is proved in Section 5.

**Theorem 2.8.** *For each manifold  $X$  there is a nontrivial functional:*

$$F : \mathcal{G}(X) \times \pi_1^{inc}(X) \rightarrow \mathbb{Q} \sqcup \{\infty\},$$

*with the following properties.*

- (1)  $F(g, \beta) \neq \infty$  if and only if  $g$  is  $\beta$ -taut.
- (2) If  $g$  is  $\beta$ -taut and  $\beta$ -regular, then it has finitely many class  $\beta$  geodesic strings, and  $F(g, \beta)$  is given by (1.2).
- (3) If  $g_0, g_1$  are joined by a  $\beta$ -taut deformation, then  $F(g_0, \beta) = F(g_1, \beta)$ .
- (4) If  $\phi : X' \rightarrow X$  is a diffeomorphism, then  $F(\phi^*g, \beta') = F(g, \phi_*\beta')$ .

For a  $\beta$ -taut  $g$ ,  $F(g, \beta)$  is defined in terms of the Fuller index.

**Corollary 2.9.** *Let  $\beta \in \pi_1^{inc}(X)$ , and let  $g_0, g_1$  be  $\beta$ -taut metrics on  $X$  with*

$$F(g_0, \beta) \neq F(g_1, \beta).$$

*Then there is no  $\beta$ -taut deformation between  $g_0$  and  $g_1$ .*

*Proof.* This is the contrapositive of Theorem 2.8(3). □

I was unable to find such a pair  $g_0, g_1$ , in fact there is strong evidence to believe the following:

**Conjecture 2.**  *$F(g, \beta)$  does not depend on the choice of a smooth structure and of a  $\beta$ -taut metric  $g$ . In other words we have the following. Let  $X$  be a topological manifold, define:*

$$\mathcal{S}(X) : \pi_1^{inc}(X) \rightarrow \mathbb{Q} \sqcup \{\infty\}$$

*by:*

$$\mathcal{S}(X)(\beta) = \begin{cases} \infty, & \text{if } X \text{ does not admit a smooth structure and a } \beta\text{-taut Finsler metric.} \\ F(g, \beta), & \text{if } g \text{ is a } \beta\text{-taut Finsler metric on } X. \end{cases},$$

*then  $\mathcal{S}$  is well-defined and hence determines a topological invariant of topological manifolds. So if  $f : X_1 \rightarrow X_2$  is a homeomorphism then  $\mathcal{S}(X_1)(\beta) = \mathcal{S}(X_2)(f_*(\beta))$ .*

Theorem 2.17 in the following section partially proves this conjecture.

*Remark 2.10.* The smooth manifold version of this conjecture, i.e. that  $\mathcal{S}(X)$  is a smooth manifold invariant, is implied by the Reeb sky catastrophe conjecture described in the next section. However, the above seems to be much more basic as will be apparent from the proof of Theorem 2.17.

### 2.0.2. Applications to sky catastrophes.

**Definition 2.11** (Preliminary). A **sky catastrophe** in class  $\beta$  for a smooth family  $\{V_t\}$ ,  $t \in [0, 1]$ , of nonvanishing vector fields on a closed manifold  $M$  is a continuous family of closed orbit strings in class  $\beta$ ,  $\tau \mapsto o_{t_\tau}$ ,  $o_{t_\tau}$  is an orbit string of  $V_{t_\tau}$ ,  $\tau \in [0, \infty)$ , such that the period of  $o_{t_\tau}$  is unbounded from above.

Sky catastrophes first appeared in the work of Fuller [22]. One general definition appears in [50], which we review in the Appendix C, the main point of this definition is that there are no regularity assumptions on  $\{V_t\}$ . The preliminary definition becomes equivalent to the more general definition given certain regularity conditions on the family  $\{V_t\}$ .

The following corollary of Theorem 2.5 may be of independent interest.

**Corollary 2.12.** *Sky catastrophes of vector fields on closed manifolds are not geodesible by metrics all of whose geodesics are minimizing. That is, if a family  $\{V_t\}$  has a sky catastrophe, there is no family  $\{g_t\}$  of metrics (all of whose geodesics are minimizing in their homotopy class) such that the orbits of  $V_t$  are unit speed parametrized  $g_t$ -geodesics.*

Fuller at the end of [23] has asked for any metric conditions on vector fields to rule out sky catastrophes. By the above, nonpositivity of curvature is one such condition. So this is a partial answer to his question.

The main conjecture in [50] is that geodesible, and more generally Reeb sky catastrophes ( $\{V_t\}$  are Reeb) are unstable, in the sense that there is a  $C^0$  nearby family  $\{V'_t\}$ , with the same endpoints, which does not have a sky catastrophe in the fixed class  $\beta$ . Let's call this **Reeb sky catastrophe conjecture**. The reason to conjecture such a thing, is that [50, Theorem 1.10] says that if  $\{o_{t_\tau}\}_\tau$  is a sky catastrophe for a family of Reeb vector fields  $\{V_t\}$  then the length of the corresponding projection  $\tau \mapsto t_\tau$  is infinite. (Assuming some minor regularity on  $\{o_{t_\tau}\}$ .) The latter suggests that the structure of such a sky catastrophe is very fragile.

*Remark 2.13.* The Reeb sky catastrophe conjecture implies the geodesible Seifert conjecture (apparently still an open problem), by the main result of [50, Theorem 1.5]. Hence, this is a subtle question.

The following is a stronger form of Corollary 2.9.

**Theorem 2.14.** *Suppose for some  $\beta$  and for a pair  $g_0, g_1$  of  $\beta$ -taut metrics on a closed manifold  $X$ :*

$$F(g_0, \beta) \neq F(g_1, \beta),$$

*then any path  $\{g_t\}$ , connecting  $g_0, g_1$ , has a sky catastrophe, meaning that the corresponding family  $\{V_t\}$  of Reeb vector fields on the unit cotangent bundle of  $X$  (cf. Section 5) has a sky catastrophe over the class  $\beta$ , in the sense of the projected form of Definition C.1 given after it.*

This is a version of [50, Theorem 3.2]; the tautness hypothesis permits a short self-contained proof, given in Appendix C. So if such a pair  $g_0, g_1$  exists, the Reeb sky catastrophe conjecture is disproved and in a rather extreme fashion.

### 2.0.3. Basic results on the invariant $F$ .

**Definition 2.15.** Let  $\beta$  be a free homotopy class of loops in  $X$ . Choose a representative of  $\beta$  and a base point  $x_0$  on its image. This choice determines an element

$$\beta_{x_0} \in \pi_1(X, x_0)$$

well-defined up to inner automorphism.

- An element  $\gamma \in \pi_1(X, x_0)$  is **indivisible** if whenever  $\gamma = \alpha^k$  for some  $\alpha \in \pi_1(X, x_0)$  and  $k > 0$ , then  $k = 1$ .
- An element  $\gamma \in \pi_1(X, x_0)$  is a  **$k$ -power** if  $\gamma = \alpha^k$ ,  $k > 1$ , for some  $\alpha$  which is not an  $n$ -power for any  $n > 1$ .
- An element  $\gamma \in \pi_1(X, x_0)$  is **atomic** if  $\gamma = \alpha^k$ , for some  $k \geq 1$ , with  $\alpha$  indivisible.

We say that the free homotopy class  $\beta$  is indivisible, respectively a  $k$ -power, respectively atomic, if  $\beta_{x_0}$  has the corresponding property for one, equivalently any, such choice of  $x_0$  and representative. This is well-defined because the corresponding notions are invariant under inner automorphism. Equivalently,  $\beta$  is a  $k$ -power precisely when it admits a smooth representative with multiplicity  $k$  whose underlying prime representative represents an indivisible class.

Note that a class  $\beta \in \pi_1^{inc}(X)$  is indivisible iff any representative of this class is prime.

*Example 5.* Let  $g$  be a Riemannian metric with negative sectional curvature on a closed manifold  $X$  and  $\beta \in \pi_1^{inc}(X)$  a class represented by a multiplicity  $n$  closed geodesic, then

$$(2.16) \quad F(g, \beta) = \frac{1}{n}.$$

In particular, if  $\beta$  is indivisible then  $F(g, \beta) = 1$ . More generally, (2.16) holds whenever  $g$  has a unique and nondegenerate geodesic string in class  $\beta$ , of even Morse index, where nondegenerate is as in the Introduction.

If  $\beta \in \pi_1^{inc}(X)$  is indivisible, then it is easy to see that the reparametrization  $S^1$  action on  $L_\beta X$  is free (Section 6), so that  $H_*^{S^1}(L_\beta X, \mathbb{Z}) \simeq H_*(L_\beta X/S^1, \mathbb{Z})$ , where  $H_*^{S^1}(L_\beta X, \mathbb{Z})$  denotes the  $S^1$ -equivariant homology. Moreover, we have the following result proved in Section 6.

**Theorem 2.17.** *Suppose that  $\beta \in \pi_1^{inc}(X)$  is indivisible, and  $X$  admits a  $\beta$ -taut metric. Then  $H_*^{S^1}(L_\beta X, \mathbb{Z})$  has finite total rank. Denote by  $\chi^{S^1}(L_\beta X)$  the Euler characteristic of this homology. Then for any  $\beta$ -taut metric  $g$  on  $X$ :*

$$F(g, \beta) = \chi^{S^1}(L_\beta X).$$

*Consequently,  $F(g, \beta)$  is independent of the choice of  $\beta$ -taut metric  $g$ , for indivisible  $\beta$ . Furthermore, Conjecture 2 holds on the subset of classes  $\beta$  which are indivisible.*

At least for surfaces  $\beta$ -taut metrics (distinct from nonpositive curvature metrics) can be constructed explicitly, recall Proposition 1.4, and Example 2 (Figure 1).

We can use the above, and the product formula of Theorem 7.1 to get:

**Theorem 2.18** (Proof in Section 8). *Every rational number has the form  $F(g, \beta)$  for some  $\beta$ -taut Riemannian  $g$  on some compact manifold  $X$  and for some  $\beta$ .*

2.0.4. *Applications to existence of negative curvature metrics.* We are going to give a certain generalization of Preissmann's theorem to fibrations, with possibly noncompact fibers, also replacing the negative sectional curvature condition by a significantly weaker condition. Though the definitions may at first seem hard to understand, we just want enough generality to incorporate interesting “flat bundle” examples (with anti-Anosov or periodic holonomy behavior). The latter are of interest in 3-manifold theory, for instance vis-à-vis Thurston's classification [51].

**Definition 2.19.** For a fiber bundle  $Z \hookrightarrow X \rightarrow Y$ , we say that  $\beta \in \pi_1^{\text{inc}}(X)$  is a **fiber class** if it lies in the image of the set map

$$i_Z : \pi_1^{\text{inc}}(Z) \rightarrow \pi_1^{\text{inc}}(X)$$

induced by inclusion of a fiber.

**Lemma 2.20.** *Let*

$$Z \hookrightarrow X \xrightarrow{p} Y$$

*be a smooth fiber bundle with connected fiber and connected base. Assume*

$$\pi_2(Y) = 0.$$

*Fix  $y_0 \in Y$ , and let*

$$H = \pi_1(Y, y_0)$$

*act on  $\pi_1^{\text{inc}}(Z_{y_0})$  by holonomy. Let  $\beta \in \pi_1^{\text{inc}}(X)$  be a fiber class, and define*

$$\mathcal{R}_{y_0}(\beta) = \{ \alpha \in \pi_1^{\text{inc}}(Z_{y_0}) \mid (i_{y_0})_*(\alpha) = \beta \}.$$

*Then  $\mathcal{R}_{y_0}(\beta)$  is a single  $H$ -orbit. In particular, if*

$$\beta_Z, \beta'_Z \in \mathcal{R}_{y_0}(\beta),$$

*then*

$$H \cdot \beta_Z = H \cdot \beta'_Z.$$

*Thus the holonomy orbit associated to the fiber class  $\beta$  is independent of the choice of fiber representative  $\beta_Z$ .*

*Proof.* Choose basepoints  $z_0 \in Z_{y_0}$  and  $x_0 = i_{y_0}(z_0)$ . Since  $\pi_2(Y) = 0$ , the homotopy long exact sequence gives an exact sequence of based groups

$$1 \rightarrow \pi_1(Z_{y_0}, z_0) \xrightarrow{(i_{y_0})_*} \pi_1(X, x_0) \xrightarrow{p_*} \pi_1(Y, y_0) \rightarrow 1.$$

Thus we identify  $\pi_1(Z_{y_0}, z_0)$  with a normal subgroup of  $\pi_1(X, x_0)$ , and the quotient is  $\pi_1(Y, y_0)$ .

Let  $\alpha, \alpha' \in \mathcal{R}_{y_0}(\beta)$ . Choose based representatives of these fiber free homotopy classes. Since they determine the same total free homotopy class  $\beta$  in  $X$ , the corresponding elements of  $\pi_1(X, x_0)$  are conjugate. Thus, after conjugating inside  $\pi_1(Z_{y_0}, z_0)$  if necessary, there exists  $x \in \pi_1(X, x_0)$  such that

$$\alpha' = x\alpha x^{-1}.$$

Let

$$h = p_*(x) \in H.$$

Conjugation by  $x$  on the normal subgroup  $\pi_1(Z_{y_0}, z_0)$  induces the holonomy action of  $h$ , up to an inner automorphism of  $\pi_1(Z_{y_0}, z_0)$ . Passing to free homotopy classes in the fiber, we get

$$\alpha' = h \cdot \alpha.$$

Thus all fiber representatives of the total free homotopy class  $\beta$  belong to a single holonomy orbit.

Conversely, if  $\alpha' = h \cdot \alpha$ , choose  $x \in \pi_1(X, x_0)$  with  $p_*(x) = h$ . Then  $x\alpha x^{-1}$  represents the same total free homotopy class in  $X$  as  $\alpha$ , and its fiber free homotopy class is  $h \cdot \alpha$ . Hence the holonomy orbit is exactly the set of fiber representatives of  $\beta$ .  $\square$

We assume from now on that  $Z, Y$  are connected. For a fiber class  $\beta \in \pi_1^{\text{inc}}(X)$ , set

$$\mathcal{R}(\beta) = \{ \alpha \in \pi_1^{\text{inc}}(Z) \mid (i_Z)_*(\alpha) = \beta \}.$$

Assume that  $\pi_2(Y) = 0$ , and put

$$H = \pi_1(Y, y_0).$$

By Lemma 2.20, the holonomy orbit

$$S = H \cdot \beta_Z$$

is independent of the choice of the fiber representative  $\beta_Z \in \mathcal{R}(\beta)$ . If another fiber inclusion is chosen, the corresponding holonomy actions are abstractly conjugate.

**Definition 2.21.** Let

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a smooth fiber bundle, with  $Y$  and  $\pi_2(Y) = 0$ , and let  $\beta \in \pi_1^{\text{inc}}(X)$  be a fiber class. We say that  $p$  is  **$\beta$ -holonomy-finite** if for one, equivalently any,  $\beta_Z \in \mathcal{R}(\beta)$ , the orbit

$$S = \pi_1(Y, y_0) \cdot \beta_Z$$

is finite.

**Definition 2.22.** Let

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a smooth fiber bundle, and let  $g$  be a Riemannian metric on  $X$ . Let

$$T^{\text{vert}}X = \ker(dp)$$

and let

$$\pi^{\text{vert}} : TX \rightarrow T^{\text{vert}}X$$

be the  $g$ -orthogonal projection. We say that  $p$  is  **$g$ -parallel** if

$$\nabla \pi^{\text{vert}} = 0.$$

Equivalently,

$$\nabla_A(\pi^{\text{vert}}B) = \pi^{\text{vert}}(\nabla_A B)$$

for all vector fields  $A, B$  on  $X$ .

**Definition 2.23.** Let  $\beta \in \pi_1^{\text{inc}}(X)$  be a fiber class, for a smooth fibration

$$p : Z \hookrightarrow (X, g) \rightarrow (Y, g_Y).$$

We say that this is a  **$\beta$ -periodic fibration** if:

- (1)  $g$  is a complete Riemannian metric.
- (2)  $Z, Y$  are connected and  $\pi_2(Y) = 0$ .
- (3)  $p$  is  $g$ -parallel and is a Riemannian submersion.
- (4)  $p$  is  $\beta$ -holonomy-finite.
- (5) the fiber metrics  $g_y$  on  $Z_y$  are taut.

The metrics  $g, g_Y$  may be kept implicit. In the above definition of a  $\beta$ -periodic fibration and the hypothesis of the following theorem we need the auxiliary metric  $g$  on  $X$  to be Riemannian, and there is no obvious extension of the theorem to the Riemann-Finsler case. However, the conclusions of the theorem are for Riemann-Finsler metrics. The following is perhaps the main technical result of the paper, proved in Section 8.

**Theorem 2.24.** *Let  $p : Z \hookrightarrow (X, g) \rightarrow (Y, g_Y)$  be a  $\beta$ -periodic fibration, where  $\beta \in \pi_1^{\text{inc}}(X)$  is a fiber class. Suppose further that  $Y$  is connected, closed,  $\chi(Y) \neq \pm 1$  and is such that all smooth closed contractible  $g_Y$ -geodesics in  $Y$  are constant. Then the following holds:*

- *$X$  does not admit a complete Riemannian metric with negative curvature.*
- *$X$  does not admit a forward complete Riemann-Finsler metric with a unique nondegenerate class  $\beta$  geodesic string.*

Note that  $\chi(Y) \neq 1$  is of course essential, as the trivial fibration  $X \rightarrow \{pt\}$ , with  $X$  admitting a complete negatively curved metric, will satisfy the hypothesis. The condition that there is a fiber class  $\beta \in \pi_1^{\text{inc}}(X)$  is also essential; as any vector bundle over a manifold admitting a Riemannian metric of negative curvature also admits a metric of negative curvature, Anderson [2].

**Corollary 2.25** (Proof in Section 8). *Let  $Y, Z$  be smooth manifolds, let  $\beta_Z \in \pi_1^{\text{inc}}(Z)$ , and assume that  $\chi(Y) \neq \pm 1$ . Suppose that  $Y$  is closed and admits a metric all of whose contractible geodesics are constant, for instance a nonpositive curvature Riemannian metric, and suppose that  $Z$  admits a taut metric. Let*

$$X = Y \times Z$$

*with projection  $X \rightarrow Y$ , let  $i : Z \rightarrow X$  be a fiber inclusion, and set*

$$\beta = i_*(\beta_Z).$$

*Then:*

- *$X$  does not admit a complete Riemannian metric of negative sectional curvature.*
- *$X$  does not admit a forward complete Riemann-Finsler metric with a unique and nondegenerate class  $\beta$  geodesic string.*

*Example 6.* Take  $Y$  any nonpositively curved closed manifold and  $Z$  a surface of genus at least one, possibly infinite type.

This theorem is very close to being sharp, for example the conclusion of the corollary is false if  $Y = S^1$  and  $Z = S^1 \times \mathbb{R}$ . As  $X = T^2 \times \mathbb{R}$  admits the warped product metric  $g_{\mathbb{R}} \times_{e^t} g_{T^2}$  (with respect to the function  $f = e^t$  on  $\mathbb{R}$ ) where  $g_{T^2}, g_{\mathbb{R}}$  are the flat metrics. This warped product has constant negative sectional curvature  $-1$ . So it is essential that not only  $\pi_1(Z)$  be nontrivial but also that there is a class incompressible to the ends. Of course,  $\chi(Y) \neq 1$  is also essential, otherwise we may take  $Y = pt$ . The condition  $\chi(Y) \neq -1$  is however not obviously essential.

A further basic set of examples for the theorem is obtained by starting with a Riemannian manifold  $(Y, g_Y)$ , and any homomorphism

$$(2.26) \quad \phi : \pi_1(Y, y_0) \rightarrow \text{Isom}(Z, g_Z), \quad (\text{the group of all isometries}).$$

where  $g_Z$  is a taut Riemannian metric and  $\beta_Z \in \pi_1^{\text{inc}}(Z)$ . (For example  $(Z, g_Z)$  is a non-simply connected complete hyperbolic surface of genus at least one). Suppose further:

- (1) The orbit

$$\text{Orb}(\beta_Z) = \{\phi_*(\gamma)(\beta_Z) \mid \gamma \in \pi_1(Y, y_0)\}$$

is finite.

- (2)  $Y$  is closed and connected.

- (3) All contractible smooth closed  $g_Y$  geodesics in  $Y$  are constant.

We have the obvious induced diagonal action

$$\pi_1(Y, y_0) \rightarrow \text{Diff}(Z \times \tilde{Y}), \text{ (the group of all diffeomorphisms),}$$

$$\gamma \mapsto ((z, y) \mapsto (\phi(\gamma)(z), \gamma \cdot y)),$$

for  $\tilde{Y}$  the universal cover of  $Y$ . Taking the quotient of  $Z \times \tilde{Y}$  by this action, we get an associated “flat” bundle  $Z \hookrightarrow X_\phi \xrightarrow{p} Y$ , with a metric  $g_\phi$  induced from the product metric  $\tilde{g} = g_Z \oplus g_Y$ , on the covering space  $q : Z \times \tilde{Y} \rightarrow Z \times Y$ . The natural projection  $p : X_\phi \rightarrow Y$  is then a Riemannian submersion and is a  $g_\phi$ -parallel. The condition on the finiteness of  $\text{Orb}$  implies that  $p$  is  $\beta$ -holonomy finite where  $\beta$  is the image by  $(i_Z)_* : \pi_1^{\text{inc}}(Z) \rightarrow \pi_1^{\text{inc}}(X_\phi)$  of  $\beta_Z$ . It follows that  $p$  is a  $\beta$ -periodic fibration. Hence  $p : (X_\phi, g_\phi) \rightarrow (Y, g_Y)$  satisfies the hypothesis of Theorem 2.24.

*Example 7.* Suppose we have  $\beta_Z \in \pi_1^{\text{inc}}(Z)$ , and let  $\phi : Z \rightarrow Z$  be an isometry of a taut metric  $g_Z$ , such that  $\text{Orb}$  is finite. Then for  $\beta = (i_Z)_*(\beta_Z)$ , by the construction above, the mapping torus

$$(Z, g_Z) \hookrightarrow (X_\phi, g_\phi) \xrightarrow{\pi} S^1$$

has the structure of a  $\beta$ -periodic fibration, satisfying the hypothesis of Theorem 2.24.

The example can be made more concrete as follows.

**Lemma 2.27.** *Let  $Z$  be a complete Riemannian manifold of finite type. Then every isometry of  $Z$  has finite orbits on  $\pi_1^{\text{inc}}(Z)$ .*

*Proof.* We first prove that, for every  $L > 0$ , only finitely many classes in  $\pi_1^{\text{inc}}(Z)$  have representatives of length at most  $L$ . Choose a compact core

$$K_0 \subset Z$$

such that  $Z - K_0$  is a finite union of collar ends. Since  $Z$  is complete, the closed  $L$ -neighborhood

$$K_L = N_L(K_0)$$

is compact. If a representative of an end-incompressible class lies entirely in  $Z - K_0$ , then it lies in a collar end and can be pushed arbitrarily far out, contradicting end-incompressibility. Therefore every representative of an end-incompressible class must meet  $K_0$ . If such a representative has length at most  $L$ , then its image lies in  $K_L = N_L(K_0)$ .

It remains to use compactness. A compact Riemannian manifold, or compact domain after a harmless thickening of  $K_L$ , contains only finitely many free homotopy classes with representatives of length at most  $L$ . For instance, choose a finite good cover with Lebesgue number  $\delta > 0$ . A loop of length at most  $L$  is encoded by an edge loop in the nerve of the cover, of combinatorial length bounded by  $N(L, \delta)$ , and there are only finitely many such combinatorial loops. Hence there are only finitely many end-incompressible classes represented by loops of length at most  $L$ .

Now let  $\phi : Z \rightarrow Z$  be an isometry and let

$$\beta_Z \in \pi_1^{\text{inc}}(Z).$$

Choose a loop  $c$  representing  $\beta_Z$ , and set

$$L = \ell(c).$$

Then  $\phi^k(c)$  represents  $\phi_*^k(\beta_Z)$  and has the same length  $L$  for every  $k \in \mathbb{Z}$ . Since  $\phi$  preserves end-incompressibility, the orbit

$$\{\phi_*^k(\beta_Z) \mid k \in \mathbb{Z}\}$$

is contained in the finite set of end-incompressible classes represented by loops of length at most  $L$ . Therefore the orbit is finite.  $\square$

**Corollary 2.28.** *Let*

$$(Z, g_Z) \hookrightarrow (X_\phi, g_\phi) \rightarrow S^1$$

*be the mapping torus of an isometry of  $Z, g_Z$ , with  $Z$  finite type and  $g_Z$  taut. Let  $\beta \in \pi_1^{\text{inc}}(X_\phi)$  be the image of some  $\beta_Z \in \pi_1^{\text{inc}}(Z)$  under fiber inclusion, as previously.*

- (1)  $X_\phi$  does not admit a complete Riemannian metric with negative curvature.
- (2)  $X_\phi$  does not admit a forward complete Riemann-Finsler metric with a unique nondegenerate class  $\beta$  geodesic string.
- (3) If  $\beta$  is indivisible,  $\chi^{S^1}(L_\beta X_\phi) = 0$  and so  $X_\phi$  does not admit a  $\beta$ -regular forward complete Finsler metric with an odd number of  $\beta$ -class geodesic strings.

The following questions seem to be open.

*Question 4.* Is the finite type assumption necessary for Corollary 2.28 in the case of surfaces? The answer is likely no, as the intuition of Nielsen-Thurston classification (it cannot be directly applied in the infinite type setting) suggests that  $\phi$  should have pseudo-Anosov type behavior which an isometry even on an infinite type surface should not have.

### 3. COMPACTNESS PRELIMINARIES

**Lemma 3.1.** *Suppose that  $g$  is a complete metric on  $X$ ,  $\beta \in \pi_1^{\text{inc}}(X)$  and let  $S \subset L_\beta X$  be a subset on which the  $g$ -length functional is bounded from above. Then the images in  $X$  of elements of  $S$  are contained in a fixed compact subset of  $X$ .*

*Proof.* Suppose otherwise. Fix an exhaustion by nested compact sets

$$\bigcup_{i \in \mathbb{N}} K_i = X, \quad K_i^\circ \supset K_{i-1}.$$

Then either there is sequence  $\{o_i\}_{i \in \mathbb{N}}$ ,  $o_i \in S$  such that  $o_i \in K_i^c$ , for  $K_i^c$  the complement of  $K_i$ , which contradicts the fact that  $\beta$  is end incompressible. Or there is a sequence  $\{o_k\}_{k \in \mathbb{N}}$ ,  $o_k \in S$  such that:

- (1) Each  $o_k$  intersects  $K_{i_0}$  for some  $i_0$  fixed.
- (2) For each  $i \in \mathbb{N}$  there is a  $k_i > i$  such that  $o_{k_i}$  is not contained in  $K_i$ .

Now if  $\text{diam}(o_k)$  is bounded in  $k$ , then condition 1 implies that  $o_k$  are contained in a set of bounded diameter. (Here  $\text{diam}(o_k)$  denotes the diameter of image  $o_k$ .) Consequently, by Hopf-Rinow theorem [8],  $o_k$  are contained in a compact set. But this contradicts condition 2, and the fact that  $K_i$  form an exhaustion of  $X$ .

Thus, we conclude that  $\text{diam}(o_k)$  is unbounded, but this contradicts the hypothesis.  $\square$

**Lemma 3.2.** *Let  $g$  be a complete metric on  $X$ , and let  $\beta \in \pi_1^{\text{inc}}(X)$ . Denote by  $S_A(g, \beta) \subset S(g, \beta)$  the subset of elements with  $g$ -length at most  $A$ . Then  $S_A(g, \beta) \subset S(g, \beta)$  is compact.*

*Proof.* By Lemma 3.1, the images of elements of  $S_A(g, \beta) \subset S(g, \beta)$  are contained in compact  $K$ . Since elements of  $S_A(g, \beta) \subset S(g, \beta)$  are parametrized by constant speed, and since we have a length bound  $S_A(g, \beta)$  is an equicontinuous collection. The result then follows by the Arzelà-Ascoli theorem and standard arguments.  $\square$

**Corollary 3.3.** *Suppose that  $g$  is a complete metric on  $X$ , all of whose class  $\beta \in \pi_1^{\text{inc}}(X)$  geodesics are minimizing, then  $g$  is  $\beta$ -taut.*

**Lemma 3.4.** *Let  $g$  be a complete metric on  $X$ , and let  $\beta \in \pi_1^{\text{inc}}(X)$ . Then there exists a closed  $g$ -geodesic minimizing length among all loops in the free homotopy class  $\beta$ .*

*Proof.* Suppose otherwise and let

$$A = \inf_{\gamma \in L_{\beta}X} \ell_g(\gamma).$$

By the hypothesis that there is no geodesic with length  $A$ , and by Lemma 3.2, there is an  $\epsilon > 0$  such that there is no closed geodesic  $o$  with  $A \leq \ell_g(o) < A + \epsilon$ .

By Lemma 3.1 the subset  $M_{\beta,A} \subset L_{\beta}X$  consisting of elements with length less than  $A + \epsilon$  is contained in a compact  $K$ . Pick  $o \in M_{\beta,A}$  and apply the curve shortening process to it. Let  $o_k, k \in \mathbb{N}$  be the stages of this process. Then  $\text{image } o_k \subset K$ . In particular  $\{o_k\}$  must converge to a geodesic, for instance [40] (we can also argue via Arzelà–Ascoli to prove convergence) but this contradicts the condition on  $\epsilon$ .  $\square$

**3.1. Proof of Theorem 2.5.** Let  $\{g_t\}, t \in [0, 1]$  be as in the hypothesis, with

$$(3.5) \quad \sup_t \left| \sup_{o \in S(g_t, \beta)} \ell_{g_t}(o) - \inf_{o \in S(g_t, \beta)} \ell_{g_t}(o) \right| < C,$$

and suppose that

$$\sup_{(o,t) \in S(\{g_t\}, \beta)} \ell_{g_t}(o) = \infty.$$

Then we have a sequence  $\{o_k\}, k \in \mathbb{N}$ , of closed class  $\beta$   $g_{t_k}$ -geodesics in  $X$ , satisfying:

- (1)  $\lim_{k \rightarrow \infty} t_k = t_{\infty} \in [0, 1]$ .
- (2)  $\lim_{k \rightarrow \infty} \ell_{g_{t_k}}(o_k) = \infty$ , where  $\ell_{g_{t_k}}(o_k)$  is the length with respect to  $g_{t_k}$ .

Applying Lemma 3.4, we may choose a length-minimizing closed  $g_{t_{\infty}}$ -geodesic  $o_{\infty}$  in the class  $\beta$ . And let  $L$  denote its  $g_{t_{\infty}}$ -length. Let  $g_{aux}$  be a fixed auxiliary metric on  $X$ , and let  $L_{aux}$  be the  $g_{aux}$  length of  $o_{\infty}$ .

Define a pseudo-metric  $d_{C^0}$  on the space of metrics on  $X$  as follows. Set  $K = \text{image } o_{\infty}$ . And set

$$V = \{v \in TX \mid \pi(v) \in K \text{ for } \pi : TX \rightarrow X \text{ the canonical projection, and } |v|_{aux} = 1\},$$

where  $|v|_{aux}$  is the norm taken with respect to  $g_{aux}$ .

Then define:

$$d_{C^0}(g_1, g_2) = \sup_{v \in V} ||v|_{g_1} - |v|_{g_2}|.$$

By Properties 1 and 2, for every  $\epsilon > 0$  we may find a  $k > 0$  such that:

$$(3.6) \quad d_{C^0}(g_{t_k}, g_{t_{\infty}}) < \epsilon$$

and

$$(3.7) \quad \ell_{g_{t_k}}(o_k) > C + L + L_{aux} \cdot \epsilon.$$

By (3.6), we have:

$$\ell_{g_{t_k}}(o_{\infty}) < \ell_{g_{t_{\infty}}}(o_{\infty}) + L_{aux} \cdot \epsilon = L + L_{aux} \cdot \epsilon.$$

Combining with (3.7) we get:

$$\ell_{g_{t_k}}(o_k) > \ell_{g_{t_k}}(o_{\infty}) + C.$$

Since we may find a closed  $g_{t_k}$ -geodesic  $o'$  satisfying  $\ell_{g_{t_k}}(o') \leq \ell_{g_{t_k}}(o_{\infty})$ , we get that

$$\left| \max_{o \in S(g_{t_k}, \beta)} \ell_{g_{t_k}}(o) - \min_{o \in S(g_{t_k}, \beta)} \ell_{g_{t_k}}(o) \right| > C,$$

and so we are in contradiction.

Thus,

$$\sup_{(o,t) \in \mathcal{O}(\{g_t\}, \beta)} \ell_{g_t}(o) < \infty.$$

Then compactness of  $\mathcal{S}(\{g_t\}, \beta)$  follows by a slight modification of the proof of Lemma 3.2.  $\square$

#### 4. PRELIMINARIES ON REEB FLOW

Let  $(C^{2n+1}, \lambda)$  be a contact manifold with  $\lambda$  a contact form, that is a one form such that  $\lambda \wedge (d\lambda)^n \neq 0$ . Denote by  $R^\lambda$  the Reeb vector field satisfying:

$$d\lambda(R^\lambda, \cdot) = 0, \quad \lambda(R^\lambda) = 1.$$

Recall that a **closed  $\lambda$ -Reeb orbit** (or just Reeb orbit when  $\lambda$  is implicit) is a smooth map

$$o : (S^1 = \mathbb{R}/\mathbb{Z}) \rightarrow C$$

such that

$$\dot{o}(t) = cR^\lambda(o(t)),$$

with  $\dot{o}(t)$  denoting the time derivative, for some  $c > 0$  called the period. Let  $S(R^\lambda, \beta)$  denote the space of all closed  $\lambda$ -Reeb orbits in free homotopy class  $\beta$ , with its compact open topology, and set

$$\mathcal{O}(R^\lambda, \beta) = S(R^\lambda, \beta)/S^1,$$

where  $S^1 = \mathbb{R}/\mathbb{Z}$  acts by reparametrization  $t \cdot o(\tau) = o(t + \tau)$ .

#### 5. DEFINITION OF THE FUNCTIONAL $F$ AND PROOFS OF AUXILIARY RESULTS

Let  $X$  be a manifold. Fix once and for all an auxiliary Riemannian metric  $g_0$ , and set

$$C = S_{g_0}^* X.$$

For any Riemann-Finsler metric  $g$ , let  $F_g^*$  denote the dual norm, and let

$$\rho_g : C \rightarrow S_g^* X, \quad \rho_g(\xi) = \frac{\xi}{F_g^*(\xi)}$$

be the fiberwise radial projection. We define a contact form on the fixed manifold  $C$  by

$$\lambda_g = \rho_g^* \lambda_{\text{can}},$$

where  $\lambda_{\text{can}}$  is the Liouville one-form on  $T^*X$ . We write

$$R^{\lambda_g}$$

for the Reeb vector field of  $\lambda_g$  on this fixed manifold  $C$ .

The same convention applies to the Sasaki metric. If  $g$  is Riemannian, let  $g_S^{\text{nat}}$  denote the usual Sasaki metric on  $S_g^* X$ , and set

$$g_S = \rho_g^* g_S^{\text{nat}}$$

on the fixed manifold  $C$ . Similarly, all metric-dependent functions on  $S_g^* X$ , such as verticality functions, are regarded as functions on  $C$  by pullback through  $\rho_g$ . This fixed- $C$  convention will be used throughout.

**5.1. Lifted geodesic strings.** We recall how closed geodesics in  $X$  are lifted to closed Reeb orbits in its cosphere bundle. Let  $C$  be the fixed cosphere model used in this paper, and let

$$q : C \rightarrow X$$

denote the projection. For a Riemann-Finsler metric  $g$ , the contact form  $\lambda_g$  on  $C$  has Reeb vector field  $R^{\lambda_g}$ . Its closed orbits project to oriented closed constant-speed  $g$ -geodesics in  $X$ .

Conversely, every oriented closed constant-speed  $g$ -geodesic has a unique Reeb lift to  $C$ . In the Riemannian case, if

$$o : S^1 \rightarrow X$$

is a nonconstant closed  $g$ -geodesic, its lift to the  $g$ -unit cotangent bundle is

$$\tilde{o}_g(t) = \frac{g_{o(t)}(\dot{o}(t), \cdot)}{|\dot{o}(t)|_g} \in S_g^*X.$$

Since  $o$  is a geodesic, this covector field is parallel along  $o$ . Thus, under the Levi-Civita splitting of  $T(T^*X)$ , the curve  $\tilde{o}_g$  is horizontal. Hence, after the constant reparametrization determined by the speed of  $o$ , it is a trajectory of the Reeb vector field. Passing to the fixed model  $C$  by radial projection gives the corresponding closed  $R^{\lambda_g}$ -orbit in  $C$ . The same statement holds in the Riemann-Finsler case using the Legendre transform.

The projection  $q$  induces a homeomorphism between closed  $R^{\lambda_g}$ -orbit strings in  $C$  and oriented closed constant-speed  $g$ -geodesic strings in  $X$ . Hence, for every class  $\beta \in \pi_1^{\text{inc}}(X)$ , if  $\mathcal{O}(g, \beta)$  denotes the class  $\beta$   $g$ -geodesic string set as usual, then its Reeb lift

$$\tilde{\mathcal{O}}(g, \beta) = \{\tilde{o} \mid o \in \mathcal{O}(g, \beta)\} \subset \mathcal{O}(R^{\lambda_g})$$

is homeomorphic to  $\mathcal{O}(g, \beta)$ . In particular, if  $\mathcal{O}(g, \beta)$  is compact, then  $\tilde{\mathcal{O}}(g, \beta)$  is compact. Moreover,  $\tilde{\mathcal{O}}(g, \beta)$  is open in  $\mathcal{O}(R^{\lambda_g})$ , since the projected free homotopy class in  $X$  is locally constant on the loop space of  $C$ .

**5.2. Definition via the Fuller index.** Suppose then that  $g$  is  $\beta$ -taut. Define

$$F(g, \beta) = i(\tilde{\mathcal{O}}(g, \beta), R^{\lambda_g}) \in \mathbb{Q},$$

where the right hand side is the Fuller index, see Appendix A. For  $g$  not  $\beta$ -taut, set  $F(g, \beta) := \infty$ .

For a path of metrics  $\{g_t\}_{t \in [0,1]}$ , the lifted cobordism of orbit string spaces

$$\tilde{\mathcal{O}}(\{g_t\}, \beta) = \{(\tilde{o}, t) \mid o \in \mathcal{O}(g_t, \beta)\}$$

is likewise homeomorphic to the downstairs cobordism  $\mathcal{O}(\{g_t\}, \beta)$ .

*Proof of Theorem 2.8.* Let  $\beta \in \pi_1^{\text{inc}}(X)$  be given.

Part (1) holds by construction.

Part (2). Let  $g$  be  $\beta$ -taut and  $\beta$ -regular. Nondegenerate geodesic strings are isolated, so the compact space  $\mathcal{O}(g, \beta)$  is finite. Then (A.3) computes  $i(\tilde{\mathcal{O}}(g, \beta), R^{\lambda_g})$  as the count (1.2).

Part (3). Let  $\{g_t\}_{t \in [0,1]}$  be a  $\beta$ -taut deformation. As discussed above, the lifted cobordism  $\tilde{\mathcal{O}}(\{g_t\}, \beta)$  is compact by  $\beta$ -tautness. It is also open in the orbit cobordism, since the projected free homotopy class in  $X$  is locally constant. Basic invariance of the Fuller index, namely (A.5), then immediately gives

$$F(g_0, \beta) = F(g_1, \beta).$$

Part (4). Given  $\beta' \in \pi_1^{\text{inc}}(X')$ , set  $\beta := \phi_*\beta'$ . The map  $\phi : (X', \phi^*g) \rightarrow (X, g)$  is an isometry. Hence it maps  $S(\phi^*g, \beta')$  homeomorphically onto  $S(g, \beta)$ , so  $\phi^*g$  is  $\beta'$ -taut if and only if  $g$  is  $\beta$ -taut, and otherwise both sides are  $\infty$ . In the taut case, the cotangent lift of  $\phi$ , composed with the radial

projections, conjugates  $R^{\lambda_{\phi^*g}}$  to  $R^{\lambda_g}$  and maps  $\tilde{\mathcal{O}}(\phi^*g, \beta')$  onto  $\tilde{\mathcal{O}}(g, \beta)$ . The Fuller index is invariant under conjugacy.  $\square$

## 6. PROOF OF THEOREM 2.17

For simplicity, as the full argument is already lengthy, we prove the theorem under the assumption that  $g$  is Riemannian. In the Finsler case we must use a more modern variational setup for the Hilbert manifold of  $H^1$ -loops. In the Finsler case the energy functional is generally only  $C^1$ , rather than  $C^2$ , on the Hilbert loop space. Nevertheless the required Palais–Smale compactness and critical point theory for closed Finsler geodesics are standard; see Caponio–Javaloyes–Masiello [19], Mercuri [38], and Lu [35]. With these replacements, the compactness argument for end-incompressible classes, and the Fuller-index invariance argument go through without substantial change.

The circle  $S^1$  acts on  $L_\beta X$  by reparametrization:

$$(s \cdot \gamma)(t) = \gamma(t + s).$$

If some  $\gamma \in L_\beta X$  had nontrivial stabilizer, then  $\gamma$  would factor through a nontrivial covering  $S^1 \rightarrow S^1$ , contradicting indivisibility. Therefore the  $S^1$ -action on  $L_\beta X$  is free. Hence

$$H_*^{S^1}(L_\beta X, \mathbb{Z}) \cong H_*(L_\beta X/S^1, \mathbb{Z}),$$

and we will construct a finite total rank model for the latter.

Let

$$\mathcal{L}_\beta X = W^{1,2}(S^1, X)_\beta$$

be the Hilbert manifold of  $H^1$ -loops in the free homotopy class  $\beta$ . Consider the energy functional

$$\mathcal{E}_g : \mathcal{L}_\beta X \rightarrow \mathbb{R}, \quad \mathcal{E}_g(\gamma) = \int_{S^1} |\dot{\gamma}(t)|_g^2 dt.$$

Since  $S(g, \beta)$  is compact,  $\mathcal{E}_g$  is bounded above on the critical set of  $\mathcal{E}_g$  in class  $\beta$ . Choose

$$E_0 > \max_{\gamma \in S(g, \beta)} \mathcal{E}_g(\gamma) = M^2,$$

which exists by the  $\beta$ -tautness assumption. Set

$$\mathcal{U}_{E_0} = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_g(\gamma) < E_0\}.$$

**Lemma 6.1.** *The inclusion*

$$\mathcal{U}_{E_0} \hookrightarrow \mathcal{L}_\beta X$$

*is an  $S^1$ -equivariant homotopy equivalence. Consequently,*

$$\mathcal{U}_{E_0}/S^1 \simeq \mathcal{L}_\beta X/S^1.$$

*Moreover, the inclusion of smooth loops into the Hilbert loop space is an  $S^1$ -equivariant homotopy equivalence, and hence*

$$\mathcal{L}_\beta X/S^1 \simeq L_\beta X/S^1.$$

*Therefore*

$$\mathcal{U}_{E_0}/S^1 \simeq L_\beta X/S^1.$$

*Proof.* The energy functional  $\mathcal{E}_g$  is smooth and  $S^1$ -invariant. We equip  $\mathcal{L}_\beta X$  with the standard  $S^1$ -invariant Hilbert metric induced by  $g$ . With respect to this metric, the negative gradient flow of  $\mathcal{E}_g$  is  $S^1$ -equivariant.

We first recall the Palais–Smale compactness needed for the deformation lemma. Let  $\{\gamma_j\} \subset \mathcal{L}_\beta X$  be a Palais–Smale sequence for  $\mathcal{E}_g$  with

$$\mathcal{E}_g(\gamma_j) \leq A.$$

Then

$$\ell_g(\gamma_j) \leq \mathcal{E}_g(\gamma_j)^{1/2} \leq A^{1/2}.$$

Since  $\beta \in \pi_1^{\text{inc}}(X)$ , Lemma 3.1 implies that the images  $\gamma_j(S^1)$  lie in a fixed compact subset of  $X$ . On this compact set, the usual Hilbert-loop compactness argument for the geodesic energy gives a strongly convergent subsequence in  $\mathcal{L}_\beta X$ . Thus  $\mathcal{E}_g$  satisfies the Palais–Smale condition on sublevels.

By the choice of  $E_0$ , the functional  $\mathcal{E}_g$  has no critical points in

$$\{\mathcal{E}_g \geq E_0\}.$$

Therefore, for every  $A > E_0$ , Palais' deformation lemma gives an  $S^1$ -equivariant homotopy equivalence

$$\{\mathcal{E}_g < E_0\} \hookrightarrow \{\mathcal{E}_g < A\}.$$

Letting  $A \rightarrow \infty$ , or equivalently applying the negative gradient flow on the exhaustion by sublevels, we obtain an  $S^1$ -equivariant homotopy equivalence

$$\mathcal{U}_{E_0} \simeq_{S^1} \mathcal{L}_\beta X.$$

Passing to quotients gives

$$\mathcal{U}_{E_0}/S^1 \simeq \mathcal{L}_\beta X/S^1.$$

Finally, the inclusion

$$C^\infty(S^1, X)_\beta \hookrightarrow W^{1,2}(S^1, X)_\beta$$

is an  $S^1$ -equivariant homotopy equivalence by standard equivariant smoothing approximation for loops. Since  $L_\beta X$  denotes the same free homotopy component in the smooth or compact-open loop space, we get

$$\mathcal{L}_\beta X \simeq_{S^1} L_\beta X.$$

Combining the quotient equivalences gives

$$\mathcal{U}_{E_0}/S^1 \simeq L_\beta X/S^1.$$

□

We now compare the Fuller index with the equivariant Morse model. The only subtlety is that after perturbing  $g$  to a regular metric  $g'$ , the full class  $\beta$  geodesic set of  $g'$  need not be compact. We therefore work in an isolating neighborhood of the compact lifted orbit block coming from  $g$ .

Let  $g_S$  denote the Sasaki metric on the fixed contact manifold  $C$  induced by  $g$  under the convention of Section 5. Define the  $S^1$ -invariant loop set  $\widehat{U}_{\sqrt{E_0}}$  to consist of all loops  $\gamma : S^1 \rightarrow C$  such that

$$[q \circ \gamma] = \beta, \quad \ell_{g_S}(\gamma) < \sqrt{E_0}.$$

Let

$$U_{\sqrt{E_0}} = \widehat{U}_{\sqrt{E_0}}/S^1$$

be its image in the orbit-string quotient. By (A.9) and the choice of  $E_0$ , this is an isolating neighborhood of

$$N_0 = \widetilde{\mathcal{O}}(g, \beta)$$

in the sense of Lemma A.6. It is also  $g_S$ -controlled, in the sense of Definition A.1, with the lower length bound supplied by the nontrivial projected class  $\beta$  and  $K$  provided by Lemma 3.1.

Let  $W$  be as in Lemma A.6, and let  $\epsilon$  be the corresponding Fuller constant for the isolating data

$$(N_0, U_{\sqrt{E_0}}, W).$$

Choose a  $\beta$ -regular complete metric  $g'$  sufficiently  $C^2$ -close  $\epsilon'$ -close to  $g$  so that, on the fixed manifold  $C$ , the vector field  $R^{\lambda_{g'}}$  is  $C^0$ -close to  $R^{\lambda_g}$ . Choose  $E'$  with

$$\max_{\gamma \in N_0} \ell_{g_S}(\gamma) < \sqrt{E'} < \sqrt{E_0}.$$

Set  $\widehat{U}$  to be the set of loops  $\gamma : S^1 \rightarrow C$  such that

$$[q \circ \gamma] = \beta, \quad \ell_{g'_S}(\gamma) < \sqrt{E'},$$

and set

$$U = \widehat{U}/S^1.$$

After shrinking the perturbation if necessary, we may assume

$$W \subset U \subset U_{\sqrt{E_0}}.$$

Thus Lemma A.8 gives

$$F(g, \beta) = i(N_1, R^{\lambda_{g'}}),$$

where

$$N_1 = U \cap \mathcal{O}(R^{\lambda_{g'}}).$$

It remains to compute the last Fuller index. Let

$$\mathcal{U}_{E'}^{g'} = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_{g'}(\gamma) < E'\}.$$

For a constant-speed  $g'$ -geodesic, its canonical lift has  $g'_S$ -length equal to its  $g'$ -length. Hence the critical circles of  $\mathcal{E}_{g'}$  in  $\mathcal{U}_{E'}^{g'}$  are in one-to-one correspondence with the elements of  $N_1$ . Since  $g'$  is  $\beta$ -regular, this critical set is a finite union of nondegenerate critical circles

$$C_o \simeq S^1.$$

Although the  $S^1$ -action on  $\mathcal{L}_\beta X$  is only continuous, each individual rotation is a smooth isometry preserving  $\mathcal{E}_{g'}$ , so the negative gradient flow of  $\mathcal{E}_{g'}$  commutes with the action, and the deformations below are  $S^1$ -equivariant. The Morse–Bott handlebody  $\mathcal{MB}$  obtained from the unstable disk bundles of these critical circles is  $S^1$ -invariant, and the standard Hilbert-manifold Morse–Bott deformation theorem gives an  $S^1$ -equivariant homotopy equivalence

$$\mathcal{MB} \simeq_{S^1} \mathcal{U}_{E'}^{g'}.$$

After passing to the quotient by  $S^1$ , the unstable disk bundle over  $C_o$  contributes one cell of dimension  $\text{morse}(o)$ ; here freeness of the action (indivisibility of  $\beta$ ) is used, as the quotient of the disk bundle by the free  $S^1$ -action is a single cell. Therefore

$$\chi(\mathcal{U}_{E'}^{g'}/S^1) = \chi(\mathcal{MB}/S^1) = \sum_{o \in N_1} (-1)^{\text{morse}(o)}.$$

We now compare the  $g'$ -sublevel with the Hilbert sublevel for  $g$ . Choose numbers  $a, b$  such that

$$M^2 < a < E' < b < E_0$$

and such that  $\mathcal{E}_g$  has no critical points on

$$\{a \leq \mathcal{E}_g \leq b\}.$$

For  $\epsilon'$  small enough we have  $\mathcal{U}_a^g \subset \mathcal{U}_{E'}^{g'} \subset \mathcal{U}_b^g$ , so that by Lemma 6.2 the inclusion

$$\mathcal{U}_{E'}^{g'} \hookrightarrow \mathcal{U}_b^g$$

is an  $S^1$ -equivariant homotopy equivalence. Since all critical points of  $\mathcal{E}_g$  have energy at most  $M^2 < b$ , the inclusion

$$\mathcal{U}_b^g \hookrightarrow \mathcal{U}_{E_0}^g$$

is also an  $S^1$ -equivariant homotopy equivalence. Thus

$$\mathcal{U}_{E'}^{g'} \simeq_{S^1} \mathcal{U}_{E_0}^g.$$

By Lemma 6.1,

$$\mathcal{U}_{E_0}^g/S^1 \simeq L_\beta X/S^1.$$

Consequently

$$\chi(\mathcal{U}_{E'}^{g'}/S^1) = \chi(L_\beta X/S^1).$$

In particular  $H_*^{S^1}(L_\beta X, \mathbb{Z})$  has finite total rank.

On the other hand, since  $g'$  is  $\beta$ -regular, the Fuller index formula, summed over the finitely many lifted free homotopy classes met by  $N_1$ , gives

$$i(N_1, R^{\lambda_{g'}}) = \sum_{o \in N_1} \frac{(-1)^{\text{morse}(o)}}{\text{mult}(o)}.$$

Since  $\beta$  is indivisible, all multiplicities in this class are 1. Hence

$$i(N_1, R^{\lambda_{g'}}) = \chi(\mathcal{U}_{E'}^{g'}/S^1) = \chi(L_\beta X/S^1).$$

Therefore

$$F(g, \beta) = \chi(L_\beta X/S^1) = \chi^{S^1}(L_\beta X).$$

□

We state the following elementary lemma separately, since it may be of independent interest.

**Lemma 6.2.** *For a metric  $h$  and a number  $r$ , set*

$$\mathcal{U}_r^h = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_h(\gamma) < r\}.$$

*Let  $a < b$ , and suppose that  $\mathcal{E}_g$  satisfies the Palais–Smale condition on the band*

$$\mathcal{B} = \{a \leq \mathcal{E}_g \leq b\}$$

*and has no critical points there. Let  $g'$  be sufficiently  $C^2$ -close to  $g$ , so that  $\nabla \mathcal{E}_{g'}$  is  $C^0$ -close to  $\nabla \mathcal{E}_g$  on  $\mathcal{B}$ . Suppose*

$$\mathcal{U}_a^g \subset \mathcal{C} \subset \mathcal{U}_b^g,$$

*where*

$$\mathcal{C} = \{\mathcal{E}_{g'} < c\}$$

*is an  $S^1$ -invariant sublevel set of  $\mathcal{E}_{g'}$ . Then the inclusion*

$$\mathcal{C} \hookrightarrow \mathcal{U}_b^g$$

*is an  $S^1$ -equivariant homotopy equivalence.*

*Proof.* By the Palais–Smale condition and the absence of critical points on  $\mathcal{B}$ , there exists  $\mu > 0$  such that

$$(6.3) \quad \|\nabla \mathcal{E}_g\| \geq \mu$$

on  $\mathcal{B}$ . For  $g'$  sufficiently close to  $g$ , we have

$$(6.4) \quad \|\nabla \mathcal{E}_{g'} - \nabla \mathcal{E}_g\| < \mu/2$$

on the same band. Therefore

$$\langle \nabla \mathcal{E}_g, \nabla \mathcal{E}_{g'} \rangle > 0$$

on  $\mathcal{B}$ .

Since

$$\mathcal{U}_a^g \subset \mathcal{C} \subset \mathcal{U}_b^g,$$

we have

$$\mathcal{U}_b^g - \mathcal{C} \subset \mathcal{U}_b^g - \mathcal{U}_a^g \subset \mathcal{B}.$$

Now along a negative  $\mathcal{E}_{g'}$ -gradient trajectory lying in  $\mathcal{B}$  we have:

$$\frac{d}{dt} \mathcal{E}_g = -\langle \nabla \mathcal{E}_g, \nabla \mathcal{E}_{g'} \rangle < 0.$$

Note that the topological boundary of  $\mathcal{U}_b^g - \mathcal{C}$  is contained in  $\{\mathcal{E}_g = b\} \cup \{\mathcal{E}_{g'} = c\}$ . Consequently the negative  $\mathcal{E}_{g'}$ -gradient flow line of an element in  $\mathcal{U}_b^g - \mathcal{C}$  cannot exit  $\mathcal{U}_b^g$  before hitting  $\mathcal{C}$ . Moreover,  $\mathcal{C}$  is forward-invariant because it is a sublevel set of  $\mathcal{E}_{g'}$ .

The estimates above also imply that  $\mathcal{E}_{g'}$  has no critical points on  $\mathcal{U}_b^g - \mathcal{C}$ . The standard deformation argument of Palais [43, Section 10, Proposition (2)], applied to  $\mathcal{E}_{g'}$  on the region  $\mathcal{U}_b^g - \mathcal{C}$ , therefore gives a deformation retraction of  $\mathcal{U}_b^g$  onto  $\mathcal{C}$ . The energies and the Hilbert metrics are  $S^1$ -invariant, so the gradient flow is  $S^1$ -equivariant. Hence the deformation retraction is  $S^1$ -equivariant.  $\square$

## 7. PRODUCT FORMULA

In this section we prove a kind of product formula for the  $F$ -count. This is used in all our applications.

**Theorem 7.1** (Product formula). *Let*

$$Z \hookrightarrow (X, g) \xrightarrow{p} (Y, g_Y)$$

*be a  $\beta$ -periodic fibration, with  $Y$  closed, connected and such that all smooth closed contractible  $g_Y$ -geodesics in  $Y$  are constant. Then  $p$  is  $\beta$ -taut. Fix  $y_0 \in Y$ , set  $Z_0 = p^{-1}(y_0)$ , and choose*

$$\beta_Z \in \pi_1^{\text{inc}}(Z_0)$$

*such that*

$$(i_{y_0})_*(\beta_Z) = \beta,$$

*where  $i_{y_0} : Z_0 \hookrightarrow X$  is the inclusion. Denote by  $g_{y_0}$  the restriction of  $g$  to the fiber of  $y_0$ . Let*

$$H = \pi_1(Y, y_0)$$

*act on  $\pi_1^{\text{inc}}(Z_{y_0})$  by holonomy, and set*

$$S_{y_0} = H \cdot \beta_Z.$$

*Then*

$$F(g, \beta) = \text{card}(S_{y_0}) \cdot \chi(Y) \cdot F(g_{y_0}, \beta_Z).$$

*Proof.* We prove  $\beta$ -tautness. We first show that every class  $\beta$   $g$ -geodesic in  $X$  is vertical. Let

$$o : S^1 \rightarrow X$$

be a closed  $g$ -geodesic representing the class  $\beta$ . Since  $\beta$  is a fiber class, the loop

$$p \circ o : S^1 \rightarrow Y$$

is contractible. Since  $p$  is a parallel Riemannian submersion, the local de Rham product decomposition for  $p$  implies that  $p \circ o$  is a  $g_Y$ -geodesic, possibly constant. Hence  $p \circ o$  is a smooth closed contractible  $g_Y$ -geodesic. By hypothesis, every such geodesic is constant. Therefore  $p \circ o$  is constant, and so  $o$  is contained in a single fiber

$$Z_y = p^{-1}(y).$$

Thus the class  $\beta$  geodesic strings in  $X$  are precisely the vertical geodesic strings whose fiber class represents  $\beta$  in  $X$ .

It remains to prove compactness. Let  $o_j$  be a sequence of class  $\beta$   $g$ -geodesic strings in  $X$ . By the previous paragraph, each  $o_j$  is contained in a fiber  $Z_{y_j}$ . Since  $Y$  is compact, after passing to a subsequence we may assume

$$y_j \rightarrow y_\infty.$$

Choose a contractible coordinate ball

$$U \subset Y$$

around  $y_\infty$ . For all large  $j$ , we have  $y_j \in U$ . Since  $p$  is a parallel Riemannian submersion, the local de Rham product theorem gives a local Riemannian product decomposition

$$p^{-1}(U) \cong Z_{y_\infty} \times U.$$

Under this identification, vertical geodesics in the fibers  $Z_{y_j}$  are identified with geodesics of the fiber metric on  $Z_{y_\infty}$ , and their fiber free homotopy classes lie in the finite holonomy orbit

$$S_{y_\infty}.$$

By the fiber-tautness condition in the definition of a  $\beta$ -periodic fibration, the union of the geodesic string sets

$$\bigcup_{\alpha \in S_{y_\infty}} S(g_{y_\infty}, \alpha)$$

is compact. Since  $S_{y_\infty}$  is finite, this is a finite union of compact sets. Therefore, after passing to a further subsequence, the vertical geodesic strings  $o_j$ , transported to the model fiber  $Z_{y_\infty}$ , converge to a fiber geodesic string. Transporting back through the local product chart gives convergence of the original  $o_j$  in the class  $\beta$  geodesic string space of  $X$ .

Thus every sequence of class  $\beta$  geodesic strings has a convergent subsequence. Hence the class  $\beta$  geodesic string set is compact, and  $g$  is  $\beta$ -taut.

Let

$$C = S^*X$$

be the unit cotangent bundle of  $(X, g)$ , with projection

$$\text{pr} : C \rightarrow X.$$

For this fixed metric calculation, this is equivalent to the fixed- $C$  convention of Section 5. Let  $R^{\lambda_g}$  denote the Reeb vector field on  $C$ . The  $F$ -invariant is computed from the compact lifted orbit set

$$N_0 = \tilde{\mathcal{O}}(g, \beta).$$

The idea is to localize contributions over critical points of a Morse function on  $Y$ .

Let

$$\pi^{\text{vert}} : TX \rightarrow T^{\text{vert}}X = \ker(dp)$$

be the  $g$ -orthogonal projection. By assumptions on  $p : X \rightarrow Y$ ,

$$\nabla \pi^{\text{vert}} = 0,$$

where  $\pi^{\text{vert}}$  is viewed as a  $(1, 1)$ -tensor field.

For  $\xi \in C$ , let

$$v_\xi \in T_{\text{pr}(\xi)}X$$

be the  $g$ -dual unit vector to  $\xi$ . Define

$$P : C \rightarrow \mathbb{R}$$

by

$$P(\xi) = |\pi^{\text{vert}} v_\xi|_g^2.$$

Thus  $P = 1$  on vertical unit covectors and  $P = 0$  on horizontal unit covectors.

Let  $f : Y \rightarrow \mathbb{R}$  be a Morse function. Define

$$\tilde{f} : C \rightarrow \mathbb{R}$$

by

$$\tilde{f} = f \circ p \circ \text{pr}.$$

Equip  $C = S^*X$  with the Sasaki metric  $g_S$  induced by  $g$ , as in the Proof of Theorem 2.17. We shall use the following two elementary identities.

First,

$$(7.2) \quad dP(R^{\lambda_g}) = 0.$$

Indeed, let  $\xi(\tau)$  be a flow line of  $R^{\lambda_g}$ , and set

$$x(\tau) = \text{pr}(\xi(\tau)).$$

Then  $x(\tau)$  is a unit-speed  $g$ -geodesic. Let

$$v(\tau) = \dot{x}(\tau) = v_{\xi(\tau)}$$

be its velocity vector field. Since  $x(\tau)$  is a geodesic,

$$\nabla_{\tau} v = 0.$$

Since  $\nabla \pi^{\text{vert}} = 0$ , we have

$$\nabla_{\tau}(\pi^{\text{vert}} v) = \pi^{\text{vert}}(\nabla_{\tau} v) = 0.$$

Therefore

$$\begin{aligned} \frac{d}{d\tau} P(\xi(\tau)) &= \frac{d}{d\tau} |\pi^{\text{vert}} v(\tau)|_g^2 \\ &= 2 \langle \nabla_{\tau}(\pi^{\text{vert}} v), \pi^{\text{vert}} v \rangle_g \\ &= 0. \end{aligned}$$

Hence  $dP(R^{\lambda_g}) = 0$ .

Second,

$$(7.3) \quad dP(\nabla_{g_S} \tilde{f}) = 0.$$

Since  $\tilde{f} = f \circ p \circ \text{pr}$  depends only on the base point  $\text{pr}(\xi)$ :

$$\nabla_{g_S} \tilde{f} \in T^{\text{hor}} C.$$

More precisely,  $\nabla_{g_S} \tilde{f}$  is the horizontal lift of

$$\nabla_g(f \circ p)$$

on  $X$ . Let  $\xi(s)$  be a horizontal curve in  $C$ , and set

$$x(s) = \text{pr}(\xi(s)).$$

By definition of the Levi-Civita horizontal distribution on  $S^*X$ , the covectors  $\xi(s)$  are parallel along  $x(s)$ . Equivalently, their  $g$ -dual vectors

$$v(s) = v_{\xi(s)}$$

are parallel along  $x(s)$ :

$$\nabla_s v(s) = 0.$$

Using again  $\nabla \pi^{\text{vert}} = 0$ , we get

$$\nabla_s(\pi^{\text{vert}} v(s)) = \pi^{\text{vert}}(\nabla_s v(s)) = 0.$$

Hence

$$\begin{aligned} \frac{d}{ds} P(\xi(s)) &= \frac{d}{ds} |\pi^{\text{vert}} v(s)|_g^2 \\ &= 2 \langle \nabla_s(\pi^{\text{vert}} v(s)), \pi^{\text{vert}} v(s) \rangle_g \\ &= 0. \end{aligned}$$

Taking  $\dot{\xi}(0) = \nabla_{g_S} \tilde{f}(\xi)$  gives

$$dP_{\xi}(\nabla_{g_S} \tilde{f}) = 0.$$

Since  $\xi$  was arbitrary,

$$dP(\nabla_{g_S} \tilde{f}) = 0.$$

Now set

$$V_{\varepsilon} = R^{\lambda_g} - \varepsilon \nabla_{g_S}(P + \tilde{f}).$$

Recall first the discussion of Fuller continuation in the Appendix. The set  $N_0 = \tilde{\mathcal{O}}(g, \beta)$  is compact by  $\beta$ -tautness. Choose a controlled isolating neighborhood  $U$  of  $N_0$  and an open neighborhood  $W$  with

$\overline{W} \subset U$ , and let  $\epsilon'$  be the corresponding Fuller constant. Take  $\epsilon > 0$  small enough so that  $V_\epsilon$  is  $C^0$   $\epsilon'$ -close to  $R^{\lambda_g}$  on the compact region determined by  $U$ . Lemma A.8 gives a compact continued set  $N_\epsilon$  and

$$F(g, \beta) = i(N_\epsilon, V_\epsilon).$$

Note that the above does not require any further properties of  $V_\epsilon$ , such as completeness. We set up the Fuller index theory to sidestep such issues.

We now identify the closed orbits in  $N_\epsilon$ . We show that  $P$  is a Lyapunov function for  $V_\epsilon$ . Mainly, along a trajectory  $\xi(\tau)$  of  $V_\epsilon$ , we have

$$\begin{aligned} \frac{d}{d\tau} P(\xi(\tau)) &= dP(R^{\lambda_g}) - \epsilon dP(\nabla_{g_S} P) - \epsilon dP(\nabla_{g_S} \tilde{f}) \\ &= -\epsilon |\nabla_{g_S} P|_{g_S}^2 \leq 0. \end{aligned}$$

Thus  $P$  is nonincreasing along the  $V_\epsilon$ -flow.

If  $\xi(\tau)$  is a closed orbit of  $V_\epsilon$ , then  $P(\xi(\tau))$  returns to its initial value after one period. Since  $P$  is nonincreasing, it must be constant along the orbit. Hence

$$\nabla_{g_S} P = 0$$

along the orbit.

On each unit cotangent sphere,  $P$  is the squared norm of the vertical component of the corresponding unit vector. Therefore the critical locus of  $P$  consists of the two critical loci

$$P = 1 \quad \text{and} \quad P = 0,$$

corresponding respectively to vertical and horizontal covectors. At  $\epsilon = 0$ , the lifted orbit set under consideration is vertical, hence lies in  $P = 1$ . Taking the controlled neighborhood  $U$  (above) so that all its elements are disjoint (as loops) from  $P = 0$ , we get that every closed orbit in  $N_\epsilon$  lies in the vertical critical locus  $P = 1$ . Consequently, every closed orbit in  $N_\epsilon$  projects to a geodesic contained in a fiber  $Z_y$ .

It remains to see over which fibers these vertical orbits occur. On  $P = 1$ , the Reeb vector field restricts to the geodesic Reeb field of the fiber  $(Z_y, g_y)$ , while  $\tilde{f}$  depends only on the base point  $y$ . The perturbation

$$-\epsilon \nabla_{g_S} \tilde{f}$$

moves the base point in the negative gradient direction of  $f$ . Therefore a vertical closed orbit of  $V_\epsilon$  can occur only over a critical point of  $f$ . Conversely, if  $y \in \text{Crit}(f)$ , then

$$\nabla_{g_S} \tilde{f} = 0$$

over  $Z_y$ , and the vertical part of  $V_\epsilon$  is just the fiber geodesic flow, up to the harmless  $P$ -term which vanishes on the vertical critical locus. Thus the closed orbits of  $V_\epsilon$  in  $N_\epsilon$  are precisely the vertical closed geodesic orbits lying in fibers over critical points of  $f$ .

We now compute the index. For  $y \in Y$ , put

$$H_y = \pi_1(Y, y),$$

acting on  $\pi_1^{\text{inc}}(Z_y)$  by holonomy. Choose

$$\beta_Z \in \pi_1^{\text{inc}}(Z_y)$$

with

$$(i_y)_*(\beta_Z) = \beta.$$

Set

$$S_y = H_y \cdot \beta_Z.$$

The Fuller index is additive over disjoint compact open subsets. Thus it remains to compute the local index contribution  $i(N_{\varepsilon,y}, V_\varepsilon)$  corresponding to the isolated subset of  $N_\varepsilon$  lying over a critical point  $y$  of  $f$ . We defer this computation to the Appendix as Lemma B.1; it yields that

$$(7.4) \quad i(N_{\varepsilon,y}, V_\varepsilon) = (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha),$$

Next, we show that  $F(g_y, \cdot)$  is constant on  $S_y$ . Indeed, let

$$\gamma : [0, 1] \rightarrow Y$$

be a path based at  $y$ . Pulling back  $p : X \rightarrow Y$  along  $\gamma$  gives a fibration over  $[0, 1]$ . By the same proof that  $p$  is  $\beta$ -taut, this pullback is a taut deformation of the fiber metric from  $(Z_y, g_y)$  back to itself, with final fiber identified with the initial one by the holonomy map  $h_\gamma : Z_y \rightarrow Z_y$ . Fuller index invariance under taut deformation gives

$$F(g_y, \alpha) = F(g_y, h_{\gamma,*}\alpha).$$

Thus  $F(g_y, \alpha)$  is constant on the holonomy orbit  $S_y$ . Hence

$$\sum_{\alpha \in S_y} F(g_y, \alpha) = \#S_y \cdot F(g_y, \beta_Z).$$

If  $y, y' \in Y$ , holonomy along any path from  $y$  to  $y'$  identifies  $S_y$  with  $S_{y'}$ . The same path gives a taut deformation between the fiber metrics, so

$$\sum_{\alpha \in S_y} F(g_y, \alpha)$$

is independent of  $y$ . So we get:

$$\begin{aligned} F(g, \beta) &= \sum_{y \in \text{Crit}(f)} i(N_{\varepsilon,y}, V_\varepsilon) \\ &= \sum_{y \in \text{Crit}(f)} (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha), \\ &= \sum_{y \in \text{Crit}(f)} (-1)^{\text{morse}(y)} \cdot \#S_{y_0} \cdot F(g_{y_0}, \beta_Z) \\ &= \chi(Y) \cdot \#S_{y_0} \cdot F(g_{y_0}, \beta_Z). \end{aligned}$$

□

**Corollary 7.5.** *Let  $X = Z \times Y$ , with  $Y$  closed, connected, such that all smooth closed contractible  $g_Y$ -geodesics in  $Y$  are constant and such that  $Z, g_Z$  is taut. Let  $\beta \in \pi_1^{\text{inc}}(X)$  be a fiber class. Then  $g = g_Z \times g_Y$  is  $\beta$ -taut. Furthermore,*

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z).$$

*Proof.* This is immediate from the theorem. □

## 8. PROOF OF THEOREM 2.24 AND ITS COROLLARIES

We prove the first part. By Cartan–Hadamard, a complete negatively curved Riemannian manifold has at most one geodesic string in each nontrivial free homotopy class; moreover the absence of nontrivial periodic Jacobi fields in negative curvature gives nondegeneracy. Existence of a closed geodesic in the end-incompressible class  $\beta$  follows from Lemma 3.4. Hence a complete negatively curved metric would give a unique nondegenerate class  $\beta$  geodesic string. It thus suffices to prove the second part.

We prove the second part. Suppose first that  $\beta$  is atomic, so that  $\beta$  has a representative that is a  $k$ -cover of a loop,  $k \geq 1$ , whose corresponding class  $\alpha$  is indivisible. By the assumption that all contractible  $g_Y$

geodesics are constant, classical Morse theory Milnor [40] tells us that  $Y$  has vanishing higher homotopy groups  $\pi_j(Y, y_0)$ ,  $j \geq 2$ . Thus by the long exact sequence of a fibration  $i_{Z,*} : \pi_1(Z, z_0) \rightarrow \pi_1(X, i_Z(z_0))$  is a group injection, and image  $i_{Z,*} = \ker p_*$ . Now  $\pi_1(Y, y_0)$  is torsion-free since it is a finite dimensional  $K(\pi_1(Y, y_0), 1)$  space, Hatcher [27, Prop. 2.45]. It follows that  $\alpha$  is also a fiber class since

$$p_*(\alpha^k) = 0 \implies p_*(\alpha) = 0.$$

Furthermore,  $\alpha \in \pi_1^{inc}(X)$  (otherwise  $\beta$  is not).

**Lemma 8.1.** *Let*

$$Z \hookrightarrow (X, g) \xrightarrow{p} (Y, g_Y)$$

*be a  $g$ -parallel Riemannian submersion with complete total space, and assume that  $Y$  is connected with  $\pi_2(Y) = 0$ . Let*

$$\beta, \gamma \in \pi_1^{inc}(X)$$

*be fiber classes such that*

$$\beta = \gamma^m$$

*for some  $m \geq 1$ . If  $g$  is  $\beta$ -taut, then  $g$  is  $\gamma$ -taut and  $p$  is  $\gamma$ -holonomy-finite.*

*Proof.* We show  $\gamma$ -tautness. Every closed  $\gamma$ -geodesic  $o_\gamma$  has  $m$ -fold cover  $o_\gamma^m \in S(g, \beta)$ , so

$$\ell_g(o_\gamma) = \frac{1}{m} \ell_g(o_\gamma^m) \leq A/m, \quad A := \max_{o \in S(g, \beta)} \ell_g(o) < \infty,$$

where finiteness of  $A$  is  $\beta$ -tautness. Hence  $S(g, \gamma) = S_{A/m}(g, \gamma)$ , which is compact by Lemma 3.2.

We show  $\gamma$ -holonomy-finiteness. Choose

$$\gamma_Z \in \mathcal{R}(\gamma)$$

and set

$$S_{y_0} = \pi_1(Y, y_0) \cdot \gamma_Z.$$

Suppose, toward a contradiction, that  $S_{y_0}$  is infinite. For each

$$\gamma_i \in S_{y_0}$$

choose a closed  $g_{y_0}$ -geodesic

$$o_i \subset Z_{y_0}$$

representing  $\gamma_i$ . Such geodesics exist by Lemma 3.4. Since the classes  $\gamma_i$  are distinct in  $Z_{y_0}$ , the loops  $o_i$  are pairwise distinct as elements of the fiber loop space.

We first record a separation property for these loops in the ambient loop space  $LX$ . Choose a contractible trivializing neighborhood

$$U \subset Y$$

of  $y_0$ . For a loop  $o \subset Z_{y_0}$ , let

$$\mathcal{V}_{U,o} \subset LX$$

be the set of loops  $\eta : S^1 \rightarrow X$  such that

$$p(\eta(S^1)) \subset U$$

and such that  $\eta$ , regarded as a loop in  $p^{-1}(U)$ , is freely homotopic in  $p^{-1}(U)$  to  $o$ . The set  $\mathcal{V}_{U,o}$  is open in  $LX$ : the condition that the image lie in  $p^{-1}(U)$  is open, and free homotopy classes are open components of the loop space of  $p^{-1}(U)$ .

Moreover, if  $o_i$  and  $o_j$  represent distinct free homotopy classes in  $Z_{y_0}$ , then

$$\mathcal{V}_{U,o_i} \cap \mathcal{V}_{U,o_j} = \emptyset.$$

Indeed, an element of the intersection would give a free homotopy between  $o_i$  and  $o_j$  inside  $p^{-1}(U)$ . Since  $U$  is contractible, the inclusion

$$Z_{y_0} \hookrightarrow p^{-1}(U)$$

is a homotopy equivalence, and hence  $o_i$  and  $o_j$  would already be freely homotopic inside  $Z_{y_0}$ , contrary to construction.

On the other hand,  $S(g, \gamma)$  is compact. Thus the infinite family  $\{o_i\} \subset S(g, \gamma)$  has an accumulation point  $o_\infty \in S(g, \gamma)$ . Since every  $o_i$  lies in the fiber  $Z_{y_0}$ , the limit  $o_\infty$  also lies in  $Z_{y_0}$ . The open neighborhood  $\mathcal{V}_{U, o_\infty}$  of  $o_\infty$  therefore contains  $o_i$  for infinitely many  $i$ , forcing these  $o_i$ 's to lie in the same fiber free homotopy class as  $o_\infty$ . This contradicts the fact that the classes  $\gamma_i$  are pairwise distinct. Hence  $S_{y_0}$  is finite and  $p$  is  $\gamma$ -holonomy-finite. □

By Theorem 7.1  $p$  is  $\beta$ -taut. By the lemma above  $p$  is  $\alpha$ -holonomy finite and hence is an  $\alpha$ -periodic fibration. Now, if  $\chi(Y) \neq \pm 1$  then by the product formula of Theorem 7.1  $F(g, \alpha) \neq \pm 1$ , since  $F(g_{y_0}, \alpha_Z)$  is an integer by Theorem 2.17.

Now, if  $X$  admits a complete metric  $h$  with a unique and nondegenerate class  $\beta$   $h$ -geodesic string  $o_\beta$ , then it clearly also has a unique class  $\alpha$   $h$ -geodesic string  $o_\alpha$ . Furthermore,  $o_\alpha$  must be nondegenerate as otherwise  $o_\beta$ , which is a  $k$ -fold cover of  $o_\alpha$  is degenerate. Then we have:

$$F(h, \alpha) = \pm 1.$$

But this is a contradiction, since  $F(h, \alpha) = F(g, \alpha)$  by Theorem 2.17.

We now prove the general case. Suppose by contradiction that  $X$  admits a complete metric  $h$  with a unique and nondegenerate class  $\beta$   $h$ -geodesic string  $o_\beta$ . By the preceding paragraph, the class  $\beta$  is not atomic. Now  $o_\beta$  covers a multiplicity one geodesic string  $o_\alpha$  in some class  $\alpha \in \pi_1^{inc}(X)$ . Moreover,  $o_\alpha$  is the unique geodesic string in its class, otherwise  $o_\beta$  would not be unique in its class. We prove that  $\alpha$  is indivisible, which will be a contradiction to  $\beta$  not being atomic and will complete the proof.

Suppose otherwise, so that  $\alpha_{x_0} = \rho^k$  for  $k > 1$  and  $\rho \in \pi_1(X, x_0)$  (with the notation of Definition 2.15). Let  $o_\rho$  be a class  $\rho$ , closed  $h$ -geodesic string (where by slight abuse  $\rho$  also denotes the class in  $\pi_1^{inc}(X)$  corresponding to the based class  $\rho$ .) It is immediate that the  $k$  cover of  $o_\rho$ ,  $o_\rho^k$  represents  $\alpha$  and is a  $h$ -geodesic string. By the uniqueness,  $o_\alpha = o_\rho^k$ . But this contradicts multiplicity one of  $o_\alpha$ . So  $\alpha$  is indivisible. □

*Proof of Corollary 2.25.* Let  $g_Z, g_Y$  be complete Riemannian metrics with  $g_Z$  taut, and  $g_Y$  not admitting contractible closed geodesics. Take the product metric  $g = g_Z \times g_Y$  on  $X = Z \times Y$ . For a class  $\beta \in \pi_1^{inc}(X)$  in the image of the inclusion  $\pi_1^{inc}(Z) \rightarrow \pi_1^{inc}(X)$ , the natural projection  $X \rightarrow Y$  is automatically a  $\beta$ -periodic fibration. Then the conclusion readily follows from Theorem 2.24. □

*Proof of Corollary 2.28.* The first two parts follow from Theorem 2.24, Lemma 2.27 and Example 7. The last part is deduced as follows. The product formula Theorem 7.1 gives us  $F(g_\phi, \beta) = 0$ . So for any  $\beta$ -regular  $g$  with finitely many class  $\beta$   $g$ -geodesic strings  $o$ , since  $\beta$  is indivisible, by Corollary 1.3 we have:

$$\chi^{S^1}(L_\beta X_\phi) = F(g, \beta) = \sum_o (-1)^{\text{morse } o} = 0.$$

Then the conclusion readily follows. □

*Proof of Theorem 2.18.* By Theorem 7.1 0 is certainly a value of the invariant  $F$ . We first prove that every negative rational number is the value of the invariant. Let  $p, q$  be positive integers. Let  $Y$  be a closed surface of genus  $(p+1) > 1$  with a hyperbolic metric  $g_Y$ , let  $Z$  be the genus 2 closed surface with a hyperbolic metric  $g_Z$  and let  $\beta_Z \in \pi_1^{inc}(Z)$  be the class represented by a  $2 \cdot q$ -fold covering of a simple closed loop representing a generator of the fundamental group of  $Z$ .

Let  $X = Y \times Z$  with the product metric  $g = g_Y \times g_Z$  and  $p : X \rightarrow Y$  the canonical projection. Taking  $\beta = (i_Z)_*(\beta_Z)$ , by Corollary 7.5,

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z) = (-2p) \cdot \frac{1}{2q} = -\frac{p}{q}.$$

So we proved our first claim.

Let again  $p, q$  be positive integers. Let  $Y$  be closed surface of genus 2, with a hyperbolic metric  $g_Y$ . And let  $Z$  be a manifold satisfying  $F(g_Z, \beta_Z) = -\frac{p}{2q}$  for some  $\beta_Z$ -taut metric  $g_Z$  on  $Z$  and for some class  $\beta_Z \in \pi_1^{inc}(Z)$ . This exists by the discussion above. Let  $g = g_Y \times g_Z$  be the product metric on  $Y \times Z$ , and  $\beta$  as above. Analogously to the discussion above we get:

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z) = (-2) \cdot \frac{-p}{2q} = \frac{p}{q}.$$

□

## 9. TAUTNESS VIA GROMOV HYPERBOLICITY

In this section we prove Proposition 2.7. We first recall the needed notions, following [15]. Let  $Y$  be a metric space, and  $I \subset \mathbb{R}$  an interval. A map  $q : I \rightarrow Y$  is a  $(\lambda, c)$ -**quasigeodesic** if for all  $s, t \in I$ :

$$\lambda^{-1}|s - t| - c \leq d(q(s), q(t)) \leq \lambda|s - t| + c.$$

It is an  $r$ -**local**  $(\lambda, c)$ -**quasigeodesic** if its restriction to every subinterval of length  $r$  is a  $(\lambda, c)$ -quasigeodesic. We use the local-to-global theorem [15, III.H.1.13]: for a  $\delta$ -hyperbolic geodesic space  $Y$  and constants  $(\lambda_0, c_0)$ , there are  $r = r(\delta, \lambda_0, c_0)$  and  $(\lambda, c) = (\lambda, c)(\delta, \lambda_0, c_0)$ , such that every  $r$ -local  $(\lambda_0, c_0)$ -quasigeodesic of  $Y$  is a  $(\lambda, c)$ -quasigeodesic.

We also use the Gronwall inequality, in the following standard form, see [25] and [26, Chapter III]. Let  $F, G$  be time dependent vector fields on an open subset of  $\mathbb{R}^m$ , with  $|F - G| \leq \epsilon'$ , and with  $F$   $\Lambda$ -Lipschitz in the space variable. Suppose  $y, z$  solve  $\dot{y} = F(t, y)$ ,  $\dot{z} = G(t, z)$ , with  $y(0) = z(0)$ . Then, while both solutions are defined,

$$(9.1) \quad |y(t) - z(t)| \leq \frac{\epsilon'}{\Lambda} (e^{\Lambda|t|} - 1).$$

Indeed  $\frac{d}{dt}|y - z| \leq |F(t, y) - F(t, z)| + |F(t, z) - G(t, z)| \leq \Lambda|y - z| + \epsilon'$ , and (9.1) solves this differential inequality.

*Proof of Proposition 2.7.* Let  $K(g_-) \leq -\kappa^2 < 0$ . The universal cover  $\tilde{X}$ , with the lift of  $g_-$ , is a  $\text{CAT}(-\kappa^2)$  space, so a  $\delta(\kappa)$ -hyperbolic geodesic space [15, III.H]. Fix  $r = r(\delta, 2, 1)$  and  $(\lambda, c)$  as in the local-to-global theorem, so both depend only on  $g_-$ .

Let  $o$  be a  $g'$ -geodesic of  $g'$ -unit speed, and  $\tilde{o}$  a lift to  $\tilde{X}$ . As  $\|g' - g_-\|_{C^1} < \epsilon$ , its  $g_-$ -speed lies in  $[1 - C\epsilon, 1 + C\epsilon]$ . Christoffel symbols involve one derivative of the metric, so  $o$  has  $g_-$ -geodesic curvature at most  $C\epsilon$ . Compare  $o$ , over a parameter arc of length  $r$ , with the  $g_-$ -geodesic sharing its initial position and velocity. Written as first order systems on  $TX$ , in charts controlled by  $g_-$ , the two solve equations as in (9.1) with  $\epsilon' = C\epsilon$  and  $\Lambda = \Lambda(g_-)$ ; so their  $C^0$ -distance over the arc is at most  $C(r)\epsilon$ . A  $g_-$ -geodesic of  $\tilde{X}$  is minimizing, and covers distance  $|s - t|$  times its speed. So on the arc, for  $\epsilon$  small,

$$\frac{1}{2}|s - t| - 1 \leq (1 - C\epsilon)|s - t| - 2C(r)\epsilon \leq d(\tilde{o}(s), \tilde{o}(t)) \leq (1 + C\epsilon)|s - t|.$$

This is where the speed bound enters, converting parameter differences to  $g_-$ -distances. So  $\tilde{o}$  is an  $r$ -local  $(2, 1)$ -quasigeodesic, and by the local-to-global theorem a  $(\lambda, c)$ -quasigeodesic.

Now let  $o$  be a class  $\beta$  closed  $g'$ -geodesic, and  $\tilde{o}$  a lift invariant under a deck transformation  $\gamma$  representing  $\beta$ . The translation length of  $\gamma$  is  $L_\beta$  — the  $g_-$ -length of the closed  $g_-$ -geodesic in class  $\beta$ . Fix  $a$  on the axis of  $\gamma$  and set  $D = d(\tilde{o}(0), a)$ . Over  $n$  periods the quasigeodesic inequality gives

$$n \ell_{g_-}(o) \leq \lambda d(\tilde{o}(0), \gamma^n \tilde{o}(0)) + \lambda c \leq \lambda n L_\beta + \lambda(2D + c),$$

and  $n \rightarrow \infty$  yields  $\ell_{g_-}(o) \leq \lambda L_\beta$ : the constant  $D$  drops out. Using the speed bound once more, in every class the geodesics of every metric in the ball have  $g'$ -lengths at most  $(1 + C\epsilon)\lambda L_\beta$ . The ball is convex, so for  $g'_0, g'_1$  in it the linear path stays in it, the length spread hypothesis of Theorem 2.5 holds in every class, and the path is a taut deformation. Theorem 2.8(3) gives the last clause.  $\square$

## 10. NEAR-FLAT TAUTNESS VIA KAM CONFINEMENT

We show that every Riemann-Finsler metric on  $T^2$  sufficiently close to a flat metric  $g_0$  is taut, with every class  $\beta$  closed geodesic of length at most twice the minimal  $g_0$ -length in the class, and is taut homotopic to  $g_0$ .

**Proposition 10.1** (Near-flat rigidity). *Let  $g_0$  be a flat Riemannian metric on  $T^2$ , and let  $L_\beta$  denote the minimal  $g_0$ -length of loops in class  $\beta$ . There are an absolute  $k_0 \in \mathbb{N}$  and a  $\delta_* = \delta_*(g_0) > 0$  with the following property. If  $g'$  is a Riemann-Finsler metric on  $T^2$  with  $\|g' - g_0\|_{C^{k_0}} < \delta_*$ , then for every nontrivial free homotopy class  $\beta$ , every closed  $g'$ -geodesic  $o$  of class  $\beta$  satisfies*

$$(10.2) \quad \ell_{g'}(o) \leq 2L_\beta.$$

Consequently  $g'$  is taut, and any two metrics in the  $\delta_*$ -ball are joined by a taut deformation; in particular every such  $g'$  is taut homotopic to  $g_0$ . Moreover, for every  $\varepsilon \in (0, 1]$  there is  $\delta(\varepsilon) \in (0, \delta_*]$  such that (10.2) holds with 2 replaced by  $1 + \varepsilon$  on the  $\delta(\varepsilon)$ -ball.

Note that  $\delta_*$  and  $\delta(\varepsilon)$  do not depend on  $\beta$ . For Finsler  $g'$ ,  $\|g' - g_0\|_{C^{k_0}}$  denotes the  $C^{k_0}$ -distance of  $F'^2$  to  $F_0^2$  on  $\{\frac{1}{2} \leq F_0 \leq 2\} \subset TT^2$ . We set  $\aleph := k_0$  and  $\hbar := \delta_*$ : these are the constants  $\aleph, \hbar$  of the Introduction.

The proof is effective: the constants  $C_1, C_2, C_3$  below are explicit, the order  $k_0$  is determined effectively by the proof of Theorem 10.3, and the threshold  $\delta_0(r, \eta)$  of Theorem 10.3 is explicitly estimable from the quantitative finite smoothness forms of the invariant curve theorem [29, 45]. So  $\delta_*$  is in principle computable. We do not pursue this here: thresholds extracted by hand from KAM iterations are far smaller than the true breakdown of invariant circles, and a realistic  $\delta_*$  would be a computer-assisted computation.

The classical theory of minimal geodesics on the torus (Morse [41], Hedlund [28], Bangert [7], Mather [37]) controls, for every metric, the *minimizers* in each class. Proposition 10.1 is orthogonal to it: the bound is on *all* class  $\beta$  closed geodesics, not just the minimizers, and no minimizer-theoretic argument reaches it. The mechanism is confinement by invariant tori of the geodesic flow which are graphs over  $T^2$  and hence, in pairs, separate the unit tangent bundle. Intrinsically this graph property is the second Birkhoff theorem, established for Lagrangian invariant tori by Bialy–Polterovich and Polterovich [14, 44, 11, 10] and, without a dynamical assumption, by Arnaud [4]; for the merely continuous invariant tori of  $T^2$  geodesic flows it is due to Bialy [12] (see also [13]). In the perturbative regime below it comes for free from KAM. The smallness of  $\delta_*$  is necessary for (10.2): the metric of Example 1 is not  $\beta$ -taut for any nontrivial  $\beta$ .

We expect taut metrics whose geodesic flow carries no incompressible invariant torus at all. Exhibiting one is a converse KAM problem, not taken up here.

In the other direction, Example 2 lies beyond the reach of the conclusion of Proposition 10.1, at every  $\delta$ . Suppose the bound (10.2) holds for  $g'$ , with 2 replaced by  $1 + \varepsilon$ , for some flat  $g_0$  with  $\|g' - g_0\|_{C^0} < \delta$ . Every class  $\beta$  curve has  $g'$ -length at least  $(1 - C\delta)L_\beta$ . So the lengths of the class  $\beta$  strings of  $g'$  spread by a factor of at most  $(1 + \varepsilon)/(1 - C\delta)$ . In the example the two class  $\beta_0$  strings have lengths  $1/2$  and  $1$ .

So the example satisfies the conclusion for no flat  $g_0$ , no  $\varepsilon < 1$  and no small  $\delta$ : it is non-perturbative for the confinement mechanism itself, not merely far from the flat metrics.

**10.1. Setup and normalization.** Every flat torus is isometric to  $\mathbb{R}^2/\Lambda$  with the Euclidean metric; composing with a linear identification  $\mathbb{R}^2/\mathbb{Z}^2 \cong \mathbb{R}^2/\Lambda$  produces a diffeomorphism  $\phi$  of  $T^2$  with  $\phi^*g_0$  constant-coefficient. Pulling the problem back by  $\phi$  distorts  $C^{k_0}$ -balls by a fixed constant and preserves geodesics, lengths and classes. So assume  $T^2 = \mathbb{R}^2/\mathbb{Z}^2$  and  $g_0 = G_0$ , a constant positive-definite matrix; to simplify notation further we take  $G_0$  to be the identity, so that  $g_0$  is the standard flat metric. The general case is identical, with the constants below depending on  $G_0$ ; see also Remark 10.6. Here and below  $C$  denotes a positive absolute constant whose value may change between occurrences, while each numbered constant  $C_1, C_2, C_3$  denotes a single specific absolute constant, produced at its first appearance and unchanged thereafter. A nontrivial free homotopy class  $\beta$  simultaneously represents a deck vector  $\beta \in \mathbb{Z}^2 - \{0\}$ , and  $L_\beta = |\beta|$ , the Euclidean norm.

The unit-speed geodesic flow of  $g'$  is the Hamiltonian flow of  $H'(x, p) = \frac{1}{2}F'^*(x, p)^2$  on  $\Sigma' = \{H' = \frac{1}{2}\} \subset T^*T^2$ , where  $F'^*$  is the dual norm; a periodic orbit of period  $T$  projects to a closed geodesic of length  $T$ . The fiberwise Legendre dual costs at most one derivative, so

$$\|H' - H\|_{C^{k_0-1}(A)} \leq C\delta \quad \text{on } A = \{\tfrac{1}{4} \leq H \leq 1\}, \quad H(p) = \tfrac{1}{2}|p|^2;$$

we rename  $k_0 - 1 \rightarrow k_0$  and, enlarging  $\delta$  by this fixed factor, assume  $\|H' - H\|_{C^{k_0}(A)} \leq \delta$ .

The flat system is integrable:  $\dot{x} = p, \dot{p} = 0$ . Its unit level  $\Sigma = \{H = \frac{1}{2}\}$  is foliated by the invariant tori  $T_\psi = \{p = (\cos \psi, \sin \psi)\}$ ,  $\psi \in S^1$ ; on  $T_\psi$  the flow is linear with frequency vector  $(\cos \psi, \sin \psi)$ . Here and for the rest of the section  $S^1 := \mathbb{R}/2\pi\mathbb{Z}$  denotes the angle circle, of total measure  $2\pi$ —in contrast with the parametrizing circle  $\mathbb{R}/\mathbb{Z}$  of loops used elsewhere in the paper. For  $\delta$  small,  $D_p^2 H'$  is close to the identity, so each locus  $\{p : H'(x, p) = \frac{1}{2}\}$  is a strictly convex curve enclosing the origin; it is therefore the radial graph of a function  $\rho(x, \cdot) : S^1 \rightarrow \mathbb{R}_{>0}$  with  $\|\rho - 1\|_{C^1} \leq C\delta$ . In particular,

$$\Psi : \Sigma' \rightarrow T^2 \times S^1, \quad (x, p) \mapsto (x, \psi := \arg p)$$

is a diffeomorphism. For  $(x, p) \in \Sigma'$ , with  $\psi = \arg p$  as above, the radial graph description of the fiber says exactly that  $p = \rho(x, \psi)(\cos \psi, \sin \psi)$ ; hence

$$|p - (\cos \psi, \sin \psi)| = |\rho(x, \psi) - 1| \leq C\delta.$$

**10.2. The KAM input.** For a real number  $r > 0$ , call  $\omega \in \mathbb{R}^2$   $(r, 2)$ -Diophantine if  $|\langle \gamma, \omega \rangle| \geq r|\gamma|^{-2}$  for every integer vector  $\gamma \in \mathbb{Z}^2 - \{0\}$ , and call  $\psi \in S^1$  *good* if  $(\cos \psi, \sin \psi)$  is  $(r, 2)$ -Diophantine. The notion of good depends on the parameter  $r$ , which the notation suppresses;  $r$  is an adjustable parameter, chosen small in the proof of Proposition 10.1.

**Theorem 10.3** (Isoenergetic KAM). *There is an absolute  $k_0$  such that for every  $r, \eta > 0$  there is  $\delta_0 = \delta_0(r, \eta) > 0$  with the following property. If  $\|H' - H\|_{C^{k_0}(A)} \leq \delta_0$ , then for every good  $\psi_0$  there is a 2-torus  $T'_{\psi_0} \subset \Sigma'$ , invariant under the  $H'$ -flow, of the form  $T'_{\psi_0} = \{(x, P(x)) : x \in T^2\}$  with  $\sup_x |P(x) - (\cos \psi_0, \sin \psi_0)| \leq \eta$ .*

This is proved by Moser's invariant curve theorem, and this route determines the absolute constant  $k_0$ : one derivative is lost passing from  $H'$  to the sector return maps below, and the remainder is the smoothness threshold of the curve theorem, fed Diophantine rotation numbers of exponent 3 — the image of the  $(r, 2)$  condition on a sector. Both inputs are explicit, so  $k_0$  is effective; we do not optimize its value. On each of four sectors of  $S^1$  on which a coordinate velocity is bounded away from zero, the flow on  $\Sigma'$  reduces to an exact area-preserving twist map of an annulus. The twist is, up to a positive factor, the bordered determinant, which here is constant:

$$\det \begin{pmatrix} D^2 H & \nabla H \\ \nabla H^\top & 0 \end{pmatrix} = -|p|^2 = -1 \quad \text{on } \{H = \tfrac{1}{2}\}.$$

For each good  $\psi_0$  the rotation number is Diophantine, by the definition of the good set. Moser's theorem [42], in Herman's finite smoothness form [29], then gives an invariant curve. Its suspension is the torus  $T'_{\psi_0}$ , and  $C^1$ -closeness to the flat section makes it a graph. For the KAM lineage of the statement—not needed for the proof—see Arnold [5] for the analytic case, Pöschel [45, 46] for quantitative versions in finite smoothness, and Broer–Huitema [16] for the isoenergetic form; the twist above is the isoenergetic nondegeneracy of  $H$ .

**Lemma 10.4** (Good directions are dense at scale  $r$ ). *There is an absolute constant  $C_1$  such that, for every  $r > 0$ ,  $\text{Leb}(S^1 - \{\text{good}\}) \leq C_1 r$ . Since the good set is closed, every complementary gap has length  $\leq C_1 r$ . In particular, when  $C_1 r < 2\pi$  the good set is nonempty and every interval in  $S^1$  of length  $> C_1 r$  contains a good angle.*

*Proof.* The good set is the complement of the union of the sets

$$B_\gamma = \{\psi \in S^1 : |\langle \gamma, (\cos \psi, \sin \psi) \rangle| < r|\gamma|^{-2}\},$$

over integer vectors  $\gamma \in \mathbb{Z}^2 - \{0\}$ . Each  $B_\gamma$  is open, so the good set is closed. Writing  $\gamma = |\gamma|(\cos \theta_\gamma, \sin \theta_\gamma)$ , we have  $\langle \gamma, (\cos \psi, \sin \psi) \rangle = |\gamma| \cos(\psi - \theta_\gamma)$ , so that

$$B_\gamma = \{\psi \in S^1 : |\cos(\psi - \theta_\gamma)| < r|\gamma|^{-3}\}.$$

For  $0 \leq s \leq 1$  the set  $\{\phi \in S^1 : |\cos \phi| < s\}$  is the union of two arcs of length  $2 \arcsin(s)$ , centered at  $\pm \pi/2$ , and  $\arcsin(s) \leq \frac{\pi}{2} s$  by convexity of  $\arcsin$  on  $[0, 1]$ ; hence

$$\text{Leb}(B_\gamma) \leq 2\pi r |\gamma|^{-3},$$

trivially also when  $r|\gamma|^{-3} > 1$ . The number of integer vectors  $\gamma$  with  $n \leq |\gamma| < n+1$  is at most  $Cn$ , so

$$\text{Leb}(S^1 - \{\text{good}\}) \leq \sum_{\gamma \neq 0} \text{Leb}(B_\gamma) \leq 2\pi r \sum_{n \geq 1} \frac{Cn}{n^3} \leq C_1 r.$$

Finally, a complementary gap is a connected component of the open set  $S^1 - \{\text{good}\}$ , so its length is at most the measure  $C_1 r$  of that set. When  $C_1 r < 2\pi$  the complement cannot be all of  $S^1$ , so the good set is nonempty, and an interval of length  $> C_1 r$  cannot lie in a single gap, so it contains a good angle.  $\square$

**Lemma 10.5** (Confinement). *For  $r$  and  $\eta$  sufficiently small and  $\delta \leq \delta_0(r, \eta)$ , along every orbit of the  $H'$ -flow on  $\Sigma'$  the momentum angle  $\psi(t)$  stays in a fixed arc of length  $W := C_2(r + \eta)$ , where one may take  $C_2 = 4C_1 + 24$ .*

*Proof.* In the coordinates  $\Psi$ , each  $T'_{\psi_0}$  is the graph of  $\Theta_{\psi_0}(x) = \arg P(x)$ , and for  $\eta \leq 1/4$  we have  $|\Theta_{\psi_0}(x) - \psi_0| \leq 4\eta$ : indeed  $|P(x)| \geq 1 - \eta$ , so the sine of the angle between  $P(x)$  and  $(\cos \psi_0, \sin \psi_0)$  is at most  $\eta/(1 - \eta)$ , and hence the angle is at most  $4\eta$ , using  $\theta \leq \frac{\pi}{2} \sin \theta$  for  $\theta \in [0, \pi/2]$ . If the orbit starts on a torus it stays there and  $\psi$  oscillates by at most  $8\eta$ . Otherwise, the two intervals below are arcs of length  $2C_1 r$ , disjoint since  $r$  and  $\eta$  are small, so by Lemma 10.4 we may choose good angles

$$\psi_1 \in (\psi(0) - 2C_1 r - 8\eta, \psi(0) - 8\eta), \quad \psi_2 \in (\psi(0) + 8\eta, \psi(0) + 2C_1 r + 8\eta);$$

then pointwise  $\Theta_{\psi_1} < \psi(0) < \Theta_{\psi_2}$  and  $\Theta_{\psi_1} < \Theta_{\psi_2}$ . The band  $R = \{(x, \psi) : \Theta_{\psi_1}(x) < \psi < \Theta_{\psi_2}(x)\}$  has  $\partial R = T'_{\psi_1} \sqcup T'_{\psi_2}$ . If the orbit ever left  $\bar{R}$ , at the first boundary time it would lie on an invariant torus, forcing the entire orbit—including its initial point—onto that torus, a contradiction. So  $\psi(t)$  lies in the arc  $[\psi_1 - 4\eta, \psi_2 + 4\eta]$ , of length at most  $4C_1 r + 24\eta \leq (4C_1 + 24)(r + \eta)$ .  $\square$

This is where two degrees of freedom enter: the tori have codimension one in  $\Sigma' \cong T^2 \times S^1$ , and a *pair* of graph-tori separates it.

### 10.3. Proof of Proposition 10.1.

*Proof. Length bound.* Let  $o$  be a closed  $g'$ -geodesic of class  $\beta$ . As a Hamiltonian trajectory on  $\Sigma'$  in the  $\Psi$  coordinates this gives a trajectory  $(x(t), p(t))$  of period  $T = \ell_{g'}(o)$ , whose lift satisfies  $\tilde{x}(T) - \tilde{x}(0) = \beta$ , so  $\frac{1}{T} \int_0^T \dot{x} dt = \beta/T$  (recalling that  $\beta$  also denotes the corresponding deck vector); in particular  $\beta \neq 0$ , as the class is nontrivial. Now

$$\dot{x}(t) = \partial_p H'(x, p) = p(t) + O(\delta) = (\cos \psi(t), \sin \psi(t)) + O(\delta),$$

using  $|p - (\cos \psi, \sin \psi)| \leq C\delta$ . By Lemma 10.5,  $\psi(t)$  lies in an arc  $I$  of length  $W$ . Set

$$e_* := \beta/|\beta|, \quad \varphi_* := \arg e_*,$$

the direction of the total displacement.

We claim  $\varphi_*$  lies within distance  $4(W + C\delta)$  of  $I$ . Indeed, fix any  $\psi^0 \in I$ ; then  $|\dot{x}(t) - (\cos \psi^0, \sin \psi^0)| \leq W + C\delta$  for every  $t$ , hence the same bound holds for the average  $\beta/T$ , and, as in the proof of Lemma 10.5, a vector within distance  $s \leq 1/4$  of a unit vector makes angle at most  $4s$  with it (smallness of  $W + C\delta$  is arranged by the choice of  $r, \eta, \delta$  below). Consequently  $|\psi(t) - \varphi_*| \leq 5W + C\delta$  for every  $t$ , and since  $\psi \mapsto (\cos \psi, \sin \psi)$  is 1-Lipschitz,

$$|\dot{x}(t) - e_*| \leq C_3(W + \delta) =: W' \quad \text{for all } t.$$

We take  $C_3 \geq 1 + C$ ; the choice of  $r, \eta, \delta$  below then gives  $W + C\delta \leq \varepsilon/4 \leq 1/4$ , as used in the claim above. Since  $|e_*| = 1$ , this gives  $\langle \dot{x}(t), e_* \rangle \geq 1 - W'$  pointwise, so that

$$T(1 - W') \leq \int_0^T \langle \dot{x}(t), e_* \rangle dt = \left\langle \int_0^T \dot{x}(t) dt, e_* \right\rangle = \langle \beta, \beta/|\beta| \rangle = |\beta| = L_\beta :$$

pairing with the direction of the displacement evaluates its length exactly, with no inequality on the right. Whence

$$T \leq L_\beta(1 - W')^{-1}.$$

Given  $\varepsilon \in (0, 1]$ , choose  $r, \eta$  with  $C_3 C_2(r + \eta) \leq \varepsilon/4$  and set  $\delta(\varepsilon) := \min\{\delta_0(r, \eta), \varepsilon/(4C_3)\}$ . For  $\delta \leq \delta(\varepsilon)$  we get  $W' \leq \varepsilon/2$ , and since  $(1 - \varepsilon/2)^{-1} \leq 1 + \varepsilon$  for  $\varepsilon \in (0, 1]$ , every class  $\beta$  closed geodesic has  $\ell_{g'}(o) \leq (1 + \varepsilon)L_\beta$ . This gives the moreover clause. Taking  $\varepsilon = 1$  and  $\delta_* := \delta(1)$  we get (10.2). The confinement constrains all unit-speed geodesics simultaneously, so this holds for every class  $\beta$  with the same  $\delta$ .

*Tautness and the deformation.* By (10.2),  $S(g', \beta) = S_{2L_\beta}(g', \beta)$ , which is compact by Lemma 3.2; so  $g'$  is taut. For  $g', g''$  in the  $\delta_*$ -ball, interpolate the squared norms,  $F_t^2 = (1 - t)F'^2 + tF''^2$ : strong convexity and 2-homogeneity are preserved, and by convexity of the ball each  $g_t$  satisfies (10.2). Hence for each  $\beta$  the length spread satisfies

$$\sup_{t \in [0, 1]} \left( \sup_{S(g_t, \beta)} \ell_{g_t} - \inf_{S(g_t, \beta)} \ell_{g_t} \right) \leq 2L_\beta < \infty,$$

and Theorem 2.5 shows  $\{g_t\}$  is  $\beta$ -taut for every  $\beta$ , i.e. a taut deformation. Taking  $g'' = g_0$  gives the last clause.  $\square$

*Remark 10.6.* Dimension two is essential and enters only in Lemma 10.5: on  $T^n$ ,  $n \geq 3$ , KAM tori have codimension  $n - 1 \geq 2$  in the energy level and separate nothing; the analogue of Proposition 10.1 is open there and adjacent to Arnold diffusion. Nothing soft can replace the confinement: pointwise bounds  $|K| \leq \delta$  do not exclude long class  $\beta$  geodesics—a  $\tau_\beta$ -periodic plane curve of geodesic curvature  $\leq \delta$  can execute U-turns at length cost  $O(1/\delta)$  apiece—so the dynamical confinement does real work. Finally, the proof used only that  $H$  is fiberwise quadratic-convex and  $x$ -independent with nonvanishing bordered determinant, so it applies verbatim to any constant-coefficient  $g_0$  (as used in the normalization above) and, more generally, to  $g_0$  any flat Minkowski (locally Minkowski Finsler) metric, whose indicatrix replaces the round circle and supplies the radial-graph property, the transversality in Lemma 10.4, and the twist.

## 11. ARITHMETIC OF CLOSED GEODESICS ON TORI

Throughout this section we use the following notation. For  $g$  a metric,  $\beta$  an indivisible class and  $d \geq 1$ ,  $P_d$  denotes the set of prime closed  $g$ -geodesic strings of class  $d\beta$ , and

$$s_d := \sum_{o \in P_d} i(o) \in \mathbb{Z}$$

its signed count, where  $i(o) := -\text{sign det}(I - \mathcal{R}_o) \in \{\pm 1\}$ , with  $\mathcal{R}_o$  the linearized return map of  $o$ . This is the local Fuller index of  $o$  in the normalization (A.2) of Appendix A, under which  $i(o) = (-1)^{\text{morse}(o)}$  for  $o$  with orientable normal bundle — automatic on an orientable surface — so that a length minimizing geodesic contributes  $+1$ . Finally,  $b_d$  denotes the number of elements of  $P_d$  of inverse-hyperbolic type ( $\mathcal{R}_o$  has negative eigenvalues).

*Proof of Theorem 1.6.* By Proposition 10.1, every  $g$  in the  $C^{k_0}$   $\delta_*$ -ball of the flat metric is  $\beta$ -taut for every nontrivial  $\beta$ , and is joined to the flat metric by a  $\beta$ -taut deformation. For the flat metric itself: after pulling back by a diffeomorphism of  $T^2$  induced by a change of basis of  $\mathbb{Z}^2$ , we may take  $\beta$  to be a multiple of the class of the first factor of  $T^2 = S^1 \times S^1$ , and  $g_0$  is joined to a product flat metric by a linear path of flat metrics, which is a taut deformation by Corollary 2.6; for the product flat metric Theorem 7.1 gives  $F = 0$ , as  $\chi(S^1) = 0$ . By Theorem 2.8(3) and (4),  $F(g, \beta) = 0$ .  $\square$

*Proof of Lemma 1.8.* Let  $a, b$  be based representatives of  $\beta, \gamma$ , so that  $xb^kx^{-1} = a^e$  for some  $x \in \pi_1(X, x_0)$ .

In the free abelian case, write  $\pi_1(X, x_0) \cong \mathbb{Z}^n \subset \mathbb{Q}^n$  additively:  $kb = ea$ , so  $b$  is a rational multiple of  $a$ . As  $\beta$  is indivisible,  $a$  is a primitive vector, and a rational multiple of a primitive vector lying in  $\mathbb{Z}^n$  is an integer multiple. So  $b = ma$ , and  $mk = e$  with  $m > 0$ .

In the torsion-free hyperbolic case, every nontrivial element  $h$  lies in a unique maximal elementary subgroup  $E(h)$ , which is infinite cyclic [15], and two maximal elementary subgroups sharing a nontrivial element coincide. As  $\beta$  is indivisible,  $a$  is primitive and  $E(a) = \langle a \rangle$ . The element  $a^e = xb^kx^{-1}$  lies in  $E(xbx^{-1})$  and in  $E(a)$ , so  $E(xbx^{-1}) = \langle a \rangle$ . Thus  $xbx^{-1} = a^m$  for some  $m \in \mathbb{Z}$ , and  $mk = e$  forces  $m > 0$ .  $\square$

**Lemma 11.1.** *Let  $X$  be a closed orientable manifold,  $\beta$  an indivisible class, and  $e \geq 1$ . Let  $g$  be an  $e\beta$ -taut and  $e\beta$ -regular metric on  $X$ , and suppose that  $\beta$  satisfies the cancellation property (Definition 1.7). Then*

$$(11.2) \quad e F(g, e\beta) = \sum_{d|e} d \sum_{o \in P_d} i(o^{e/d}),$$

with  $P_d$  the set of prime geodesic strings of class  $d\beta$  and  $i$  the local Fuller index (A.2).

*Proof.* Tautness and regularity of the divisor classes  $d\beta$  are automatic: covers of strings are strings, and an eigenvalue 1 of  $\mathcal{R}_o$  is one of  $\mathcal{R}_{o^{e/d}}$ . In particular each  $P_d$ ,  $d|e$ , is finite. Every class  $e\beta$  geodesic string is the  $k$ -fold cover  $o^k$  of a unique prime geodesic string  $o$ , and  $o^k$  has multiplicity  $k$ . The class  $\gamma$  of  $o$  satisfies  $k\gamma = e\beta$ , so by hypothesis  $o \in P_d$  with  $dk = e$ . So the class  $e\beta$  strings are in bijection with the pairs  $(d, o)$ , where  $d|e$  and  $o \in P_d$ , the pair corresponding to  $o^{e/d}$ , of multiplicity  $e/d$ . By the definition of  $F$  and Appendix A, the contribution of  $o^{e/d}$  to  $F(g, e\beta)$  is its local Fuller index divided by its multiplicity, that is  $i(o^{e/d})/(e/d)$ . Summing over the pairs and multiplying by  $e$  gives (11.2), since  $e/(e/d) = d$ .  $\square$

**Lemma 11.3.** *Let  $o$  be a closed geodesic string of a complete Riemann-Finsler metric on a manifold, with linearized return map  $\mathcal{R} = \mathcal{R}_o$ , and let  $j \geq 1$  be such that  $o$  and  $o^j$  are nondegenerate. If  $j$  is odd then  $i(o^j) = i(o)$ . If  $j$  is even and  $o^2$  is nondegenerate then  $i(o^j) = i(o^2)$ .*

*Proof.*  $\det(I - \mathcal{R}^j) = \prod_{\lambda} (1 - \lambda^j)$ , over the eigenvalues of  $\mathcal{R}$  with multiplicity, and the eigenvalue multiset is invariant under  $\lambda \mapsto \lambda^{-1}$  and under conjugation. A pair  $\{\lambda, \lambda^{-1}\}$ ,  $\lambda > 1$ , contributes  $(1 - \lambda^j)(1 - \lambda^{-j}) < 0$  for every  $j$ . A pair  $\{-\lambda, -\lambda^{-1}\}$ ,  $\lambda \geq 1$ , contributes a positive factor for  $j$  odd and a negative one for  $j$  even, when nonzero. A pair  $\{e^{\pm i\theta}\}$  contributes  $|1 - e^{ij\theta}|^2 > 0$  when nonzero. A quadruple  $\{\lambda, \bar{\lambda}, \lambda^{-1}, \bar{\lambda}^{-1}\}$  off the real and unit loci contributes  $|1 - \lambda^j|^2 |1 - \lambda^{-j}|^2 > 0$ . The factors are nonzero exactly at the nondegenerate iterates. So  $\text{sign } \det(I - \mathcal{R}^j)$  depends only on the parity of  $j$ , where defined, and by (A.2) so does  $i(o^j)$ .  $\square$

*Proof of Theorem 1.9.* The hypotheses of Lemma 11.1 hold at every divisor level  $e'|e$ , tautness and regularity being automatic. For  $d|e'$  the exponent  $e'/d$  is odd, and the iterates  $o^{e'/d}$ ,  $o \in P_d$ , are nondegenerate by  $e'\beta$ -regularity; so Lemma 11.3 gives  $\sum_{o \in P_d} i(o^{e'/d}) = s_d$ , and (11.2) at level  $e'$  reads

$$e' F(g, e'\beta) = \sum_{d|e'} d s_d.$$

This is (1.10), and Möbius inversion over the divisors of  $e$  gives (1.11). If  $F(g, d\beta) = 0$  for every  $d|e$ , then (1.11) gives  $s_e = 0$ . If  $F(g, d\beta) = 1/d$  for every  $d|e$ , then (1.11) gives

$$s_e = \frac{1}{e} \sum_{d|e} \mu(e/d) = 0$$

for  $e > 1$ . In both cases each prime geodesic string contributes  $\pm 1$  to  $s_e$ , so  $\#P_e$  is even.  $\square$

**Lemma 11.4.** *Let  $o$  be a prime closed geodesic of a Riemann-Finsler metric on a closed surface whose iterates  $o^k$ ,  $k \leq K$ , are all nondegenerate, with linearized return map  $\mathcal{R} = \mathcal{R}_o \in \text{Sp}(2, \mathbb{R})$ . Then  $\mathcal{R}$  is positive hyperbolic (eigenvalues  $\lambda^{\pm 1}$ ,  $\lambda > 1$ ), inverse-hyperbolic (eigenvalues  $-\lambda^{\pm 1}$ ,  $\lambda > 1$ ) or elliptic (eigenvalues  $e^{\pm i\theta}$ ), and for  $k \leq K$ :*

$$\text{sign } \det(I - \mathcal{R}^k) = \begin{cases} -1 & \mathcal{R} \text{ positive hyperbolic,} \\ +1 & \mathcal{R} \text{ elliptic,} \\ (-1)^{k+1} & \mathcal{R} \text{ inverse-hyperbolic.} \end{cases}$$

*Proof.* Nondegeneracy of  $o$  and  $o^2$  excludes the eigenvalue pairs  $\{1, 1\}$  and  $\{-1, -1\}$ , giving the trichotomy. If the eigenvalues are  $\lambda^{\pm 1}$ ,  $\lambda > 1$ , then

$$\det(I - \mathcal{R}^k) = (1 - \lambda^k)(1 - \lambda^{-k}) < 0.$$

If they are  $e^{\pm i\theta}$ , then

$$\det(I - \mathcal{R}^k) = |1 - e^{ik\theta}|^2 > 0,$$

nonzero by nondegeneracy of  $o^k$ . If they are  $-\lambda^{\pm 1}$ ,  $\lambda > 1$ , then

$$\det(I - \mathcal{R}^k) = (1 - (-\lambda)^k)(1 - (-\lambda)^{-k}),$$

positive for  $k$  odd and negative for  $k$  even.  $\square$

The vanishing of Theorem 1.6 has a complete arithmetic expression in terms of prime geodesic strings. A Finsler metric that is  $\beta$ -regular for all  $\beta$  is  $C^\infty$ -generic by the bumpy metric theorem, in the Riemannian case due to Abraham [1] and Anosov [3] and in the Finsler case to Rademacher–Taimanov [49]; for such  $g$ , in each taut class the geodesic strings are isolated points of a compact set, hence finite in number. Fix an indivisible class  $\beta_0$ ; the notation of the section preamble applies with  $\beta = \beta_0$ .

**Proposition 11.5.** *Let  $g$  be a Riemann-Finsler metric on  $T^2$ ,  $\beta_0$  an indivisible class, and  $e \geq 1$ . Suppose  $g$  is  $e\beta_0$ -taut and  $e\beta_0$ -regular, and  $F(g, d\beta_0) = 0$  for every divisor  $d$  of  $e$  — as holds for every  $e$  for any  $d\beta_0$ -regular metric  $C^{k_0}$ -close to a flat metric, by Theorem 1.6. Then*

$$(11.6) \quad s_e = -b_{e/2},$$

where  $b_{e/2} := 0$  for  $e$  odd. Conversely, for  $g$   $d\beta_0$ -taut and  $d\beta_0$ -regular: if (11.6) holds for every divisor  $e$  of  $d$ , then  $F(g, d\beta_0) = 0$ .

Tautness and regularity of the divisor classes are automatic, as in the proof of Lemma 11.1. In particular the counts and  $F$ -values in the hypothesis are defined. The content of Proposition 11.5 lies in the divisible classes, which have no simple representatives, and where the minimizers, as well as whole sweepout families, may consist of iterates of class  $\beta_0$  geodesics; the prime counts are then invisible to such methods, which do not track isotropy.

*Proof of Proposition 11.5.*  $\pi_1(T^2) = \mathbb{Z}^2$  is free abelian, so  $\beta_0$  satisfies the cancellation property (Lemma 1.8) and Lemma 11.1 applies at every divisor level (tautness and regularity being automatic, as noted above), giving

$$(11.7) \quad dF(g, d\beta_0) = \sum_{e|d} e \sum_{o \in P_e} i(o^{d/e}).$$

Now let  $u_e$  be the number of positive hyperbolic minus the number of elliptic elements of  $P_e$ , so that  $s_e = u_e - b_e$  and, by Lemma 11.4,  $\sum_{o \in P_e} i(o^k) = u_e + (-1)^k b_e$  for every  $k$ . Since  $(-1)^{d/e} = 1$  exactly when  $d/e$  is even, that is when  $e$  divides  $d/2$ , substitution in (11.7) gives

$$dF(g, d\beta_0) = \sum_{e|d} e(u_e - b_e) + 2 \sum_{e|d/2} e b_e = \sum_{e|d} e s_e + 2 \sum_{e|d/2} e b_e,$$

the second sum being empty for  $d$  odd. With  $B_x := \sum_{e|x} e b_e$ , and  $B_x := 0$  when  $x$  is not an integer, the vanishing of  $F(g, d\beta_0)$  for all  $d$  reads

$$(11.8) \quad \sum_{e|d} e s_e = -2B_{d/2} \quad \text{for all } d \geq 1.$$

Now for any  $d$ ,

$$(11.9) \quad \sum_{e|d} e b_{e/2} = \sum_{2f|d} 2f b_f = 2B_{d/2},$$

since the terms with  $e$  odd vanish, and the even divisors of  $d$  are exactly  $e = 2f$  with  $f$  a divisor of  $d/2$ . This gives the converse direction at once: if  $s_e = -b_{e/2}$  for every  $e$ , then the left hand side of (11.8) is the negative of that of (11.9), so (11.8) holds. For the forward direction argue by induction on  $d$ , the case  $d = 1$  of (11.8) being  $s_1 = -2B_{1/2} = 0$ . Assuming  $s_e = -b_{e/2}$  for every proper divisor  $e$  of  $d$ , (11.8) and (11.9) give

$$d s_d = -2B_{d/2} + \sum_{e|d, e \neq d} e b_{e/2} = -2B_{d/2} + (2B_{d/2} - d b_{d/2}) = -d b_{d/2},$$

hence  $s_d = -b_{d/2}$ . □

Proposition 11.5 is not accessible to  $S^1$ -equivariant Morse homology techniques, as the latter does not suitably track isotropy information. Nevertheless, there is a family resemblance with the resonance identities of closed geodesic and closed characteristic theory, where finite collections of periodic objects satisfy identities involving average Euler characteristics and mean indices; see, for example, [48, 30] and the later noncontractible resonance identities [33]. The constraints of the present paper are of a different character: the resonance identities are real valued relations driven by the asymptotics of the index, while ours are exact integer identities among the signed counts of prime geodesic strings, driven by the vanishing of a rational count.

*Proof of Theorem 1.1.* Regularity of the divisor classes of  $e\beta_0$  follows from  $e\beta_0$ -regularity in both cases, as noted after Proposition 11.5.

In the near flat case, by Theorem 1.6,  $g$  is  $d\beta_0$ -taut and  $F(g, d\beta_0) = 0$  for every  $d \geq 1$ . So Proposition 11.5 applies, giving (11.6). For  $e$  odd this reads  $s_e = 0$ ; each prime geodesic string contributes  $\pm 1$ , so their number is even. For  $e = 2n$  it reads  $s_{2n} = -b_n$ , so  $\#P_{2n} \geq |s_{2n}| = b_n$ .

In the negative curvature case,  $g$  is taut and  $F(g, d\beta_0) = F(g_-, d\beta_0)$  for every  $d$ , by Proposition 2.7. The unique class  $d\beta_0$  closed  $g_-$ -geodesic is the  $d$ -fold cover of the prime one in class  $\beta_0$ , so  $F(g_-, d\beta_0) = 1/d$  by Example 5. The class  $\beta_0$  satisfies the cancellation property,  $\pi_1(X)$  being torsion-free hyperbolic (Lemma 1.8), and Lemma 11.4 applies. So the derivation of (11.8) in the proof of Proposition 11.5 goes through with  $mF(g, m\beta_0) = 1$  in place of 0, giving

$$\sum_{d|m} d s_d = 1 - 2B_{m/2} \quad \text{for every divisor } m \text{ of } e.$$

This system determines  $s_d$ ,  $d|e$ , recursively from the  $b_d$ , and  $s_1 = 1$ ,  $s_d = -b_{d/2}$  for  $d > 1$  solves it, by (11.9). So for  $e$  odd,  $s_e = 0$  when  $e > 1$ , giving the pairing, while  $s_1 = 1$  makes  $\#P_1$  odd; and for  $e = 2n$ ,  $\#P_{2n} \geq |s_{2n}| = b_n$  as before.  $\square$

## 12. EXAMPLES

*Proof of Proposition 1.4.* The momenta  $p = a(y)\dot{x}$  are conserved, and for a unit speed geodesic

$$\dot{y}^2 + V_p(y) = 1,$$

where

$$V_p(y) := \langle p, a(y)^{-1}p \rangle.$$

A class  $\beta$  geodesic has zero  $y$ -winding: the lift of its  $S^1$ -coordinate satisfies  $\int_0^\ell \dot{y} dt = 0$ , for  $\ell$  the length. So  $\dot{y}$  vanishes somewhere, and there  $V_p = 1$ , by unit speed. If moreover  $V'_p = 0$  at that point, it is an equilibrium of the  $y$ -dynamics  $\ddot{y} = -\frac{1}{2}V'_p(y)$ , so by ODE uniqueness  $\dot{y}$  vanishes everywhere: the geodesic is a parallel over a critical point of  $V_p$  at level 1. Otherwise it is a turning point.

Either way  $V_p = 1$  somewhere, so  $|p| \geq c > 0$ , with  $c$  depending only on  $a$ . Write  $\beta = (m, 0)$  as in the statement of the proposition: the lift of the  $x$ -coordinate to  $\mathbb{R}^{n-1}$  advances by  $m$  over the period, that is  $\int_0^\ell \dot{x} dt = m$ . Now  $\dot{x} = a(y)^{-1}p$ , and the eigenvalues of  $a^{-1}$  are bounded below by  $\lambda > 0$ ; so over the full length  $\ell$ ,

$$|m| = \left| \int_0^\ell \dot{x} dt \right| \geq \lambda \ell |p| \geq \lambda c \ell,$$

so no cancellation in the advance is possible, and the length is bounded in class  $\beta$ . Tautness follows by Lemma 3.2. For the deformation, interpolate  $a_t(y) = (1-t)a(y) + t\bar{a}$ , with  $\bar{a}$  the constant matrix splitting  $T^n = T^{n-1} \times S^1$  as a Riemannian product. Now, each  $a_t$  is Clairaut, the constants above are uniform in  $t$ , so the space of pairs  $(o, t)$ , with  $o$  a class  $\beta$   $g_t$ -geodesic, is compact — a  $\beta$ -taut deformation. For the product flat metric, Corollary 7.5 with  $Y = S^1$  gives  $F = 0$ , as  $\chi(S^1) = 0$ ; so  $F(g, \beta) = 0$  by Theorem 2.8(3).  $\square$

**12.1. A lone prime geodesic string on  $T^2$ .** We now give the details of Example 2; the situation is illustrated in Figure 1. Take  $T^2 = \mathbb{R}^2/\mathbb{Z}^2$  and let  $\beta_0$  be the indivisible class of the circle  $\{y = 0\}$ .

Recall the metric of the example: in Fermi form  $g = dy^2 + A(x, y)^2 dx^2$ , with

$$A(x, y) = 1 - \frac{1}{2} \left( 1 + \frac{5}{\pi^2} \cos 2\pi x \right) \sin^2 \pi y.$$

Then  $A > 0$ , and along the curve  $o = \{y = 0\}$ :

$$K(x) = \pi^2 + 5 \cos 2\pi x.$$

We write  $o$  also for the class  $\beta_0$  geodesic string with this image. The linearized return map  $\mathcal{R}_o$  is the monodromy of the Hill equation over one period. So  $o$  is inverse-hyperbolic exactly when  $\text{tr } \mathcal{R}_o < -2$ :

the eigenvalues are then  $-\mu^{\pm 1}$ ,  $\mu > 1$ . For constant  $K \equiv c^2$  the monodromy is conjugate to the rotation by angle  $c$ , with  $\text{tr} = 2 \cos c$ ; so the boundary value  $\text{tr} = -2$  is first attained at  $K \equiv \pi^2$ .

By the oscillation theorem [36, Chapter II, Theorem 2.1], the instability intervals of the Hill equation have endpoints the periodic and antiperiodic eigenvalues; the antiperiodic tongue emanating from this constant value is opened at leading order by the first Fourier harmonic of  $K$  [32], see also [6]. This dictates the parameters above. A numerical evaluation of the Hill discriminant gives  $\text{tr} \mathcal{R}_o = -2.158 \dots < -2$ . So  $o$  is inverse-hyperbolic, and with eigenvalues  $-\mu^{\pm 1}$ ,

$$\det(I - \mathcal{R}_o) = (1 + \mu)(1 + \mu^{-1}) > 0, \quad \det(I - \mathcal{R}_o^2) = (1 - \mu^2)(1 - \mu^{-2}) < 0,$$

so  $i(o) = -1$  while  $i(o^2) = +1$ : the contribution flips on doubling.

The curve  $\{y = 1/2\}$  likewise gives a class  $\beta_0$  geodesic string, of length  $\int_0^1 A(x, 1/2) dx = 1/2$ , and the curvature along it is negative, so it is positive hyperbolic, with  $i = +1$ . A numerical computation of the return map to  $\{x = 0\}$  finds the period doubled class  $2\beta_0$  prime nondegenerate geodesic string  $\gamma$ , of length  $1.969 \dots$  and elliptic type ( $\text{tr} \mathcal{R}_\gamma = -0.492 \dots$ ), and no further class  $\beta_0$  or  $2\beta_0$  geodesic strings among the geodesics winding monotonically in the  $x$  direction; the completeness of this census beyond the monotone window, and the tautness of the two classes, we have verified only numerically.

#### APPENDIX A. FULLER INDEX AND SKY CATASTROPHES

In this appendix we recall the part of Fuller's periodic orbit index used in the paper, and fix the notation for orbit strings and for continuation under small perturbations of vector fields. Some of this material overlaps with [50]; we nevertheless give a brief self-contained account, especially since the noncompact setting needed here is not treated there, and the details serve to fix notation used throughout the paper.

Let  $X$  be a smooth manifold, and fix a complete auxiliary Riemannian metric  $h$  on  $X$ . Let

$$\mathcal{V}(X)$$

denote the space of smooth vector fields on  $X$ , with the  $C^0$ -topology on compact subsets.

Let

$$LX = C^\infty(S^1, X)$$

denote the smooth free loop space. For  $V \in \mathcal{V}(X)$ , define

$$S(V) = \{o \in LX \mid o \text{ is nonconstant, } \dot{o}(t) = T V(o(t)) \text{ for some } T > 0\}.$$

We call it the period of  $o$ , and write

$$T = T(o),$$

it is clearly uniquely determined.

The circle  $S^1 = \mathbb{R}/\mathbb{Z}$  acts on  $S(V)$  by reparametrization:

$$(s \cdot o)(t) = o(t + s).$$

We define the space of orbit strings by

$$\mathcal{O}(V) = S(V)/S^1.$$

If  $\beta$  is a free homotopy class in  $X$ , we write

$$S(V, \beta) \subset S(V), \quad \mathcal{O}(V, \beta) \subset \mathcal{O}(V)$$

for the corresponding subspaces. When no confusion is possible, we use the same letter  $o$  for a parametrized orbit and for its associated orbit string.

Let

$$q : LX \rightarrow LX/S^1$$

be the quotient map. If

$$U \subset LX/S^1,$$

we write

$$\widehat{U} = q^{-1}(U) \subset LX.$$

**Definition A.1.** We say that  $U$  is *h-controlled* if there are constants

$$0 < \ell_- < \ell_+ < \infty$$

and a compact set  $K \subset X$  such that every loop  $o \in \widehat{U}$  satisfies

$$\text{image}(o) \subset K, \quad \ell_- \leq \ell_h(o) \leq \ell_+.$$

The lower bound excludes collapse to constant loops; in the applications it is usually supplied by a nontrivial projected free homotopy class.

For an orbit string  $o \in \mathcal{O}(V)$ , its multiplicity

$$\text{mult}(o) \in \mathbb{N}$$

is the unique positive integer such that  $o$  is the  $\text{mult}(o)$ -fold cover of a prime orbit string. Equivalently, if  $T(o)$  is the period of  $o$  and  $T_{\text{prim}}(o)$  is the period of the underlying prime orbit string, then

$$\text{mult}(o) = \frac{T(o)}{T_{\text{prim}}(o)}.$$

**A.1. Fuller index.** Suppose

$$N \subset \mathcal{O}(V)$$

is finite and consists of nondegenerate orbit strings. Fuller associates to  $(N, V)$  the rational number

$$i(N, V) = \sum_{o \in N} \frac{i(o)}{\text{mult}(o)}.$$

Here, for a nondegenerate orbit string  $o$  of the flow of  $V$ , on an ambient manifold of dimension  $2m+1$ , and with linearized return map  $\mathcal{R}_o$  computed on a local hypersurface transverse to the image of  $o$ , we set

$$(A.2) \quad i(o) := (-1)^m \text{sign det}(I - \mathcal{R}_o).$$

Up to the dimensional sign  $(-1)^m$  this is the fixed point index of the time- $T(o)$  return map; the sign is chosen so that minimizing closed geodesics contribute positively, see below.

By a compact open subset  $N$  of an orbit-string space, we mean a subset which is compact and open in the relative topology. In this case Fuller again associates to this a rational index  $i(N, V)$ . The latter can be defined perturbatively, and using the formula as above.

In the special case of a nondegenerate closed geodesic orbit, with ambient manifold  $S_g X$  so that  $m = \dim X - 1$ , the Morse index theorem gives the classical relation  $\text{sign det}(I - \mathcal{R}_o) = (-1)^{\text{morse}(o) + m}$ ,<sup>4</sup> see for example [34, Chapter 12], so that by (A.2):

$$i(o) = (-1)^{\text{morse}(o)}.$$

Consequently, if  $N$  consists of finitely many nondegenerate closed orbits of the geodesic Reeb flow  $R^{\lambda_g}$  on  $C$ , we get:

$$(A.3) \quad i(N, R^{\lambda_g}) = \sum_{o \in N} \frac{(-1)^{\text{morse}(o)}}{\text{mult}(o)}.$$

---

<sup>4</sup> $\text{morse}(o)$  refers to the Morse index of a representative of the class  $o$ .

A.2. **Continuation.** Let

$$\{V_t\}_{t \in [0,1]} \subset \mathcal{V}(X)$$

be a continuous family. Define

$$S(\{V_t\}) = \{(o, t) \in LX \times [0, 1] \mid o \in S(V_t)\},$$

and

$$(A.4) \quad \mathcal{O}(\{V_t\}) = S(\{V_t\})/S^1,$$

where  $S^1$  acts on the loop coordinate by reparametrization as before.

Let

$$\mathcal{N} \subset \mathcal{O}(\{V_t\})$$

be compact and open. Suppose that its endpoint slices are

$$N_0 = \mathcal{N} \cap \mathcal{O}(V_0), \quad N_1 = \mathcal{N} \cap \mathcal{O}(V_1).$$

Then

$$(A.5) \quad i(N_0, V_0) = i(N_1, V_1).$$

We refer to this as the **basic invariance** of the Fuller index.

For applications, we need a local version of this statement under small perturbations. This is a variation of [23, Lemma 4.1], with the added control needed for noncompact  $X$ .

**Lemma A.6** (Isolation lemma). *Let  $h$  be a complete auxiliary Riemannian metric on  $X$ . Let  $V_0 \in \mathcal{V}(X)$ , be a nonsingular vector field, and let*

$$N_0 \subset \mathcal{O}(V_0)$$

*be compact and open. Let*

$$U \subset LX/S^1$$

*be an  $h$ -controlled open neighborhood of  $N_0$  such that*

$$\overline{U} \cap \mathcal{O}(V_0) = N_0.$$

*Let  $W$  be an open neighborhood of  $N_0$  satisfying*

$$\overline{W} \subset U.$$

*Then there exists  $\epsilon > 0$  with the following property. If  $V$  is a smooth vector field satisfying*

$$\sup_{x \in K} |V(x) - V_0(x)|_h < \epsilon,$$

*where  $K$  is the compact set from the  $h$ -control of  $U$ , then*

$$\mathcal{O}(V) \cap U \subset W.$$

*In particular,*

$$\mathcal{O}(V) \cap U$$

*is compact and open in  $\mathcal{O}(V)$ .*

*Proof.* Let  $K$ ,  $\ell_-$ , and  $\ell_+$  be the compact set and length bounds from the  $h$ -control of  $U$ . Since  $V_0$  is nonsingular and  $K$  is compact, there are constants

$$0 < B < A < \infty$$

such that

$$B \leq |V_0(x)|_h \leq A$$

for all  $x \in K$ . We choose  $\epsilon < B/2$ . Then every vector field  $V$  satisfying

$$\sup_K |V - V_0|_h < \epsilon$$

obeys

$$|V(x)|_h \geq B/2, \quad |V(x)|_h \leq A + \epsilon$$

for all  $x \in K$ .

Suppose the conclusion failed for the chosen  $W$ . Then there would be vector fields

$$V_j \rightarrow V_0$$

uniformly on  $K$ , and orbit strings

$$o_j \in (\mathcal{O}(V_j) \cap U) - W.$$

Choose parametrized representatives, still denoted  $o_j$ , satisfying

$$(A.7) \quad \dot{o}_j(s) = T_j V_j(o_j(s)).$$

Since  $o_j \in \widehat{U}$ , controlledness gives

$$\text{image}(o_j) \subset K, \quad \ell_- \leq \ell_h(o_j) \leq \ell_+.$$

For  $j$  sufficiently large,

$$B/2 \leq |V_j|_h \leq A + \epsilon$$

on  $K$ . Hence

$$\frac{\ell_-}{A + \epsilon} \leq T_j \leq \frac{2\ell_+}{B}.$$

The velocities are also uniformly bounded:

$$|\dot{o}_j(s)|_h = T_j |V_j(o_j(s))|_h \leq \frac{2\ell_+(A + \epsilon)}{B}.$$

Thus the  $o_j$  are equicontinuous and have images in the fixed compact set  $K$ . By the Arzelà–Ascoli theorem, after passing to a subsequence,

$$o_j \rightarrow o_\infty$$

uniformly. After passing to a further subsequence,  $T_j \rightarrow T_\infty$ , with  $T_\infty > 0$ .

Hence  $o_j$  is a Cauchy sequence in the  $C^1$  space of maps, by the defining equation (A.7). Then  $C^1$  completeness gives that

$$\dot{o}_\infty = T_\infty V_0(o_\infty).$$

Thus  $o_\infty$  is a closed  $V_0$ -orbit string. Moreover

$$o_\infty \in \overline{U} - W.$$

This contradicts

$$\overline{U} \cap \mathcal{O}(V_0) = N_0 \subset W.$$

Therefore

$$\mathcal{O}(V) \cap U \subset W$$

for all  $V$  sufficiently close to  $V_0$  on  $K$ .

It remains to prove compactness. Let  $o_j$  be any sequence in  $\mathcal{O}(V) \cap U$ . The same estimates as above give uniformly bounded periods and velocities, with all images contained in  $K$ . Arzelà–Ascoli therefore gives a convergent subsequence. Its limit is again a closed  $V$ -orbit. Since the inclusion just proved gives

$$\mathcal{O}(V) \cap U = \mathcal{O}(V) \cap \overline{W}$$

and  $\overline{W} \subset U$ , the limit lies in  $\mathcal{O}(V) \cap U$ . Thus  $\mathcal{O}(V) \cap U$  is compact. It is open because  $U$  is open.  $\square$

We call such an  $\epsilon$  a **Fuller constant** for the data

$$(V_0, N_0, U, W).$$

It also depends on the auxiliary control data  $K, \ell_-, \ell_+$ , which will always be understood from the chosen  $U$ .

**Lemma A.8** (Fuller continuation). *Let  $V_0$ ,  $N_0$ ,  $U$ ,  $W$ , and  $\epsilon$  be as in Lemma A.6. Let*

$$\{V_t\}_{t \in [0,1]}, \quad V_{t=0} = V_0$$

*be a continuous family such that for every  $t \in [0, 1]$ ,*

$$\sup_{x \in K} |V_t(x) - V_0(x)|_h < \epsilon,$$

*where  $K$  is the compact set from the  $h$ -control of  $U$ . Define*

$$\mathcal{N} = (U \times [0, 1]) \cap \mathcal{O}(\{V_t\}).$$

*Then  $\mathcal{N}$  is compact and open in the orbit cobordism  $\mathcal{O}(\{V_t\})$ . Its endpoint slices are*

$$N_0 = U \cap \mathcal{O}(V_0), \quad N_1 = U \cap \mathcal{O}(V_1).$$

*Consequently,*

$$i(N_0, V_0) = i(N_1, V_1).$$

*We call  $N_1$  the Fuller continuation of  $N_0$ .*

*Proof.* Since  $U$  is open,

$$\mathcal{N} = (U \times [0, 1]) \cap \mathcal{O}(\{V_t\})$$

is open in  $\mathcal{O}(\{V_t\})$ .

By Lemma A.6, applied to each  $V_t$ , we have

$$\mathcal{O}(V_t) \cap U \subset W$$

for every  $t \in [0, 1]$ . Since

$$\overline{W} \subset U,$$

this implies

$$\mathcal{N} = (\overline{W} \times [0, 1]) \cap \mathcal{O}(\{V_t\}).$$

The compactness then works as in Lemma A.6. The endpoint slices are

$$\mathcal{N} \cap \mathcal{O}(V_0) = U \cap \mathcal{O}(V_0) = N_0$$

and

$$\mathcal{N} \cap \mathcal{O}(V_1) = U \cap \mathcal{O}(V_1) = N_1.$$

Fuller's basic invariance applied to the compact open cobordism  $\mathcal{N}$  gives

$$i(N_0, V_0) = i(N_1, V_1).$$

□

*Example 8* (Geodesic flows and the Sasaki metric). Let  $(Y, g_Y)$  be a complete Riemannian manifold, and let

$$C_g = S^*Y$$

be its unit cotangent bundle with Liouville contact form  $\lambda_{g_Y}$ . The Reeb vector field  $R^{\lambda_{g_Y}}$  is the geodesic Reeb field.

Let  $g_S$  be the Sasaki metric on  $S^*Y$ . It is defined using the Levi-Civita connection of  $g_Y$ . Thus

$$TC = T^{\text{vert}}C \oplus T^{\text{hor}}C$$

is an orthogonal splitting, where

$$T^{\text{vert}}C = \ker(\text{pr}_*)$$

and  $T^{\text{hor}}C$  is the Levi-Civita horizontal subbundle. If  $\xi, \eta \in T(S^*Y)$  are horizontal, then

$$g_S(\xi, \eta) = g_Y(\text{pr}_* \xi, \text{pr}_* \eta),$$

and the vertical part is measured using the metric induced by  $g_Y$  on the cotangent fibers.

The basic property needed in the paper is the following. If  $o$  is a nonzero constant-speed  $g_Y$ -geodesic and  $\tilde{o}$  is its unit cotangent lift, then

$$(A.9) \quad \ell_{g_S}(\tilde{o}) = \ell_{g_Y}(o).$$

Indeed, the covector component of  $\tilde{o}$  is parallel along  $o$ , so the vertical part of  $\dot{\tilde{o}}$  vanishes and the Sasaki speed equals the base speed.

If  $\beta \in \pi_1^{\text{inc}}(Y)$  and  $U \subset S(g_Y, \beta)$  is  $g_Y$ -length bounded, then the set of canonical lifts

$$\tilde{U} \subset S(R^{\lambda_{g_Y}})$$

is  $g_S$ -length bounded by (A.9).

## APPENDIX B. A LOCAL FIBERED CALCULATION OF $F$

We prove in this section the technical lemma computing the local index, used in Theorem 7.1.

Let

$$P : S^*X \rightarrow \mathbb{R}$$

be the verticality function as in the proof of Theorem 7.1,

$$P(\xi) = |\pi^{\text{vert}} v_\xi|_g^2.$$

Here  $v_\xi$  is the  $g$ -dual unit vector to  $\xi$ , and

$$\pi^{\text{vert}} : TX \rightarrow T^{\text{vert}}X$$

is the  $g$ -orthogonal projection. The notation  $\beta$  and  $\beta_Z$  is likewise as in Theorem 7.1.

**Lemma B.1** (Local product calculation). *Let*

$$Z \hookrightarrow X \xrightarrow{p} Y$$

*be a parallel Riemannian submersion. Let*

$$\tilde{f} = f \circ p \circ \text{pr}$$

*for a Morse function  $f : Y \rightarrow \mathbb{R}$ , and define the vector field on  $S^*X$*

$$V_\varepsilon = R^{\lambda_g} - \varepsilon \nabla_{g_S}(P + \tilde{f}).$$

*Fix*

$$y \in \text{Crit}(f),$$

*and let*

$$N_{\varepsilon, y}$$

*be the isolated set of closed  $V_\varepsilon$ -orbit strings lying over the fiber*

$$Z_y = p^{-1}(y),$$

*as in the proof of Theorem 7.1. Then*

$$i(N_{\varepsilon, y}, V_\varepsilon) = (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha),$$

*where  $S_y$  is the  $\pi_1(Y, y)$ -orbit of the corresponding fiber representative in  $\pi_1^{\text{inc}}(Z_y)$ . In particular, if the fibration is holonomy-finite in the relevant class, the sum is finite.*

*Proof.* Let

$$T^{\text{vert}}X = \ker(dp), \quad H = (T^{\text{vert}}X)^{\perp_g}.$$

Since the submersion is parallel, the two distributions

$$T^{\text{vert}}X \quad \text{and} \quad H$$

are preserved by the Levi-Civita connection. Since  $T^{\text{vert}}X$  and  $H = (T^{\text{vert}}X)^{\perp_g}$  are parallel complementary distributions, the local de Rham decomposition theorem gives a product decomposition

$$p^{-1}(D_y) \cong Z_y \times D_y, \quad g = g_y \oplus g_Y|_{D_y},$$

for a normal ball neighborhood  $D_y \subset Y$  centered at  $y$ . See Kobayashi–Nomizu [31, Chapter IV].

For this local calculation we suppress the fixed- $C$  identification and write  $S^*X$  for the  $g$ -unit cotangent bundle. Let

$$C^{\text{vert}} = \{P = 1\} \subset S^*X$$

be the vertical locus. Over  $Z_y$ , this locus is naturally identified with

$$S^*Z_y \subset S^*X.$$

Near

$$S^*Z_y \subset C^{\text{vert}},$$

the product decomposition gives the normal-bundle model

$$(B.2) \quad S^*Z_y \times D_y \times B_y^*,$$

where

$$B_y^* \subset T_y^*Y$$

is a small ball in the horizontal cotangent direction. In this model, the vertical locus is

$$S^*Z_y \times D_y \times \{0\}.$$

A covector near the vertical locus decomposes as

$$\xi = \xi_{\text{vert}} + \eta, \quad \eta \in B_y^*.$$

The unit condition is

$$|\xi_{\text{vert}}|_{g_y^*}^2 + |\eta|_{g_Y^*}^2 = 1.$$

Therefore

$$P(\xi) = |\xi_{\text{vert}}|_{g_y^*}^2 = 1 - |\eta|_{g_Y^*}^2.$$

Thus  $P$  has a Morse–Bott maximum along  $C^{\text{vert}}$ .

In the local model (B.2),

$$N_{\varepsilon, y} = N_y^{\text{fib}} \times \{y\} \times \{0\},$$

where  $N_y^{\text{fib}}$  is the relevant closed orbit set of the fiber geodesic Reeb flow on  $S^*Z_y$ .

By a  $C^\infty$ -small perturbation of  $g_y$ , under which both sides of the claimed identity are unchanged by the invariance of the Fuller index, we may assume the fiber strings are nondegenerate, and hence finite in number.

We now describe the local return map at an element of  $N_{\varepsilon, y}$ . Along the vertical locus over  $Z_y$ , the fiber component of  $V_\varepsilon$  is the geodesic Reeb field  $R^{\lambda_{g_y}}$  on  $S^*Z_y$ . The base component is

$$-\varepsilon \nabla_{g_Y} f$$

on  $D_y$ . The normal covector component is the outward radial component coming from

$$-\varepsilon \nabla_{g_S} P$$

on  $B_y^*$ .

Let

$$o \in N_y^{\text{fib}},$$

and choose a product transverse section

$$\Sigma = \Sigma^{\text{fib}} \times D_y \times B_y^*,$$

where

$$\Sigma^{\text{fib}} \subset S^*Z_y$$

is transverse to the fiber Reeb flow near  $o$ , and let  $T(o)$  be the return time. In the coordinates (fiber, base, normal) of the product section, denote by  $\sigma$  the local return map of  $V_\varepsilon$ ; it is block triangular over the normal variable  $\eta$ . The normal component is a map  $\kappa$  of  $B_y^*$  alone, the time- $T(o)$  map of the radial field induced by  $-\varepsilon \nabla_{g_S} P$ , while the fiber and base components depend on  $\eta$  but restrict at  $\eta = 0$  to, respectively, the local return map  $\Phi_{\text{fib}}$  of the fiber geodesic flow generated by  $R^{\lambda_{g_y}}$ , and the time- $T(o)$  map  $\psi_y$  of  $-\varepsilon \nabla_{g_Y} f$  near  $y$ .

The fixed point of  $\sigma$  corresponding to  $o$  is  $(w, y, 0)$ , where  $w$  is the point where  $o$  meets the fiber section. Consequently the linearized return map  $\mathcal{R}_o = d\sigma$  at this fixed point is block triangular with diagonal blocks  $d\Phi_{\text{fib}}$ ,  $d\psi_y$  and  $d\kappa$ , so that

$$\det(I - \mathcal{R}_o) = \det(I - d\Phi_{\text{fib}}) \cdot \det(I - d\psi_y) \cdot \det(I - d\kappa);$$

compare also the multiplicativity of the fixed point index under products, Dold [21, Chapter VII, Section 5]. Here, for a map  $\phi$  with an isolated fixed point  $x$ , we write  $i_{\text{fp}}(\phi, x) \in \mathbb{Z}$  for its fixed point index: the local degree at  $x$  of the map  $\text{id} - \phi$ , computed in any chart; at a nondegenerate fixed point,

$$i_{\text{fp}}(\phi, x) = \text{sign } \det(I - d\phi_x).$$

Hence

$$i_{\text{fp}}(\sigma, (w, y, 0)) = i_{\text{fp}}(\Phi_{\text{fib}}, w) \cdot i_{\text{fp}}(\psi_y, y) \cdot i_{\text{fp}}(\kappa, 0).$$

In normal coordinates at  $y$ , the map  $\text{id} - \psi_y$  equals  $\varepsilon T(o) \nabla_{g_Y} f$  to first order, so has the same local degree as  $\nabla_{g_Y} f$ . Thus, because  $y$  is a Morse nondegenerate critical point,

$$i_{\text{fp}}(\psi_y, y) = (-1)^{\text{morse}(y)}.$$

Since the normal component of

$$-\varepsilon \nabla_{g_S} P$$

has outward radial linearization on a vector space of dimension  $\dim Y$ , the map  $\text{id} - \kappa$  has local degree  $(-1)^{\dim Y}$ . Hence

$$i_{\text{fp}}(\kappa, 0) = (-1)^{\dim Y}.$$

Therefore

$$i_{\text{fp}}(\sigma, (w, y, 0)) = (-1)^{\dim Y + \text{morse}(y)} i_{\text{fp}}(\Phi_{\text{fib}}, w).$$

The multiplicity of  $o$  is unchanged by taking the product with the isolated base and normal fixed points. Finally we pass from fixed point indices to the Fuller weights of (A.2): the ambient manifolds are  $S^*X$  on the left and  $S^*Z_y$  on the right, of dimensions  $2(\dim Z + \dim Y) - 1$  and  $2\dim Z - 1$ , so the two dimensional signs differ by exactly  $(-1)^{\dim Y}$ , cancelling the factor above:

$$\frac{i(o; V_\varepsilon)}{\text{mult}(o)} = (-1)^{\text{morse}(y)} \frac{i(o; R^{\lambda_{g_y}})}{\text{mult}(o)}.$$

Summing over all fiber orbit strings in the local block gives

$$i(N_{\varepsilon, y}, V_\varepsilon) = (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha).$$

This proves the claim.  $\square$

## APPENDIX C. SKY CATASTROPHES

We now recall the topological formulation of sky catastrophes as studied in [50]. This is the formulation necessary for Section 2.0.2.

**Definition C.1.** Let

$$\{V_t\}_{t \in [0,1]} \subset \mathcal{V}(X)$$

be a continuous family. We say that the family has a sky catastrophe in class  $\beta$  if there is an endpoint orbit string

$$y \in \mathcal{O}(V_0, \beta) \cup \mathcal{O}(V_1, \beta) \subset \mathcal{O}(\{V_t\}, \beta)$$

such that no compact open subset of  $\mathcal{O}(\{V_t\}, \beta)$  contains  $y$ .

Equivalently, the endpoint orbit string  $y$  cannot be included in any compact continuation block through the family.

For suitably regular families, this agrees with the preliminary Definition 2.11.

Now let  $\{V_t\}$  be a family of Reeb vector fields on the fixed cosphere model  $C$  of Section 5, and  $\beta \in \pi_1^{\text{inc}}(X)$ . Denote by  $\mathcal{O}_\beta(\{V_t\}) \subset \mathcal{O}(\{V_t\})$  the subspace of orbit strings whose projection to  $X$  has class  $\beta$ . We say that  $\{V_t\}$  has a **sky catastrophe over the class  $\beta$**  if some endpoint orbit string in  $\mathcal{O}_\beta(\{V_t\})$  is contained in no compact open subset of  $\mathcal{O}_\beta(\{V_t\})$ . This projected form is the correct one for lifts: on a surface the class of the lift of a class  $\beta$  string is not unique, being shifted by the fiber class according to a turning number, while in dimension at least 3 the fiber of  $C \rightarrow X$  is simply connected, and a sky catastrophe over  $\beta$  is one in the single lifted class.

*Proof of Theorem 2.14.* Let  $\{V_t\}$  be the family of Reeb vector fields on  $C$  induced by  $\{g_t\}$ . Suppose, for contradiction, that  $\{V_t\}$  has no sky catastrophe over the class  $\beta$ . The endpoint slices of  $\mathcal{O}_\beta(\{V_t\})$  are the lift sets  $\tilde{\mathcal{O}}(g_i, \beta)$ ,  $i = 0, 1$ , compact by  $\beta$ -tautness and the lift homeomorphism of Section 5.1. By hypothesis, every  $y$  in these slices lies in a compact open subset  $C_y$  of  $\mathcal{O}_\beta(\{V_t\})$ . By compactness, finitely many  $C_y$  cover the two slices; let  $\mathcal{N}$  be their union: compact, and open in  $\mathcal{O}(\{V_t\})$ , as the projected free homotopy class is locally constant on the loop space of  $C$  (Section 5.1). Each endpoint slice  $\mathcal{N} \cap \mathcal{O}(V_i)$  contains all of  $\tilde{\mathcal{O}}(g_i, \beta)$ , and nothing else, so equals it. The basic invariance (A.5) then gives

$$F(g_0, \beta) = i(\tilde{\mathcal{O}}(g_0, \beta), V_0) = i(\tilde{\mathcal{O}}(g_1, \beta), V_1) = F(g_1, \beta),$$

contradicting the hypothesis.  $\square$

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